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Generalized Arithmetic Functions

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Generalized Arithmetic Functions

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Abstract

Generalized Arithmetic Functions

Jonathan Cass

In this paper we seek to generalize σ , the sum-of-divisors function and Euler's totient function, ϕ . These functions, σ and ϕ , are two of the most well-known arithmetic functions and have been studied extensively. Once we have defined the generalizations, we study some of the properties (in particular, growth rates, multiplicativity, and fixed points) of these generalized functions as well as their products, convolutions, and compositions. It has been noted that there is a dual nature to σ and ϕ , and we find that this extends well to the generalized versions through the use of conjugate pairs.

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Chapter 1

Introduction

Number theory is often thought of as the study of the natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$. Despite this seemingly simple subject, modern number theory uses very advanced machinery - from the complex functions of analytic number theory to the number fields in algebraic number theory. In the beginning of number theory, no such tools existed and the study of $\mathbb{N}$ was generally through so-called arithmetic functions, i.e. functions $f : \mathbb{N} \rightarrow \mathbb{N}$. We define some of the most commonly studied functions now, and introduce some notation.

$$\begin{aligned} U(n) &= 1 && \text{for all } n \\ I(n) &= n && \text{for all } n \\ \mu(p_1^{e_1} \dots p_l^{e_l}) &= \begin{cases} (-1)^l & \text{if } e_i = 1 \text{ for all } i \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

$$\begin{aligned}\sigma(n) &= \sum_{d|n} d \\ \phi(n) &= \#\{1 \leq k \leq n \mid \gcd(k, n) = 1\}\end{aligned}$$

Indeed, it turns out that $\sigma = I * U$ and $\phi = I * \mu$ (cf. Proposition 2.2.5), where $*$ is the Dirichlet convolution (cf. pg. 16). In this dissertation, we develop a theory of generalized arithmetic functions.

We generalize arithmetic functions by defining new functions for any $P, Q \subset \mathbb{N}$, as follows:

$$\begin{aligned}U_P(n) &= \begin{cases} 1 & n \in P \\ 0 & n \notin P \end{cases} \\ \mu_Q(n) &= \sum_{\substack{d|n \\ d \in Q}} \mu\left(\frac{n}{d}\right)\end{aligned}$$

Then as special cases we get

$$\begin{aligned}\mu_{\{1\}} &= \mu & U_{\{1\}} &= E \\ \mu_{\mathbb{N}} &= E & U_{\mathbb{N}} &= U\end{aligned}$$

and to generalize ϕ and σ we write

$$\phi_Q = I * \mu_Q \qquad \sigma_P = I * U_P$$

so that

$$\phi_{\{1\}} = I * \mu_{\{1\}} = I * \mu = \phi$$

$$\phi_{\mathbb{N}} = I * \mu_{\mathbb{N}} = I * E = I$$

$$\sigma_{\{1\}} = I * U_{\{1\}} = I * E = I$$

$$\sigma_{\mathbb{N}} = I * U_{\mathbb{N}} = I * U = \sigma$$

We continue now with two definitions that will be used throughout.

Definition Let $P \subset \mathbb{N}$. Then we say that P is a factor set if, given relatively prime $r, s \in \mathbb{N}$, we have $rs \in P$ if and only if $r \in P$ and $s \in P$.

Definition Suppose that P and Q are factor sets, and that given $n \in \mathbb{N}$ there exist unique $a \in P$, $b \in Q$ such that $n = ab$. Then we say that (P, Q) is a conjugate pair.

Some examples of conjugate pairs are $(\mathbb{N}, \{1\})$, $(\mathbb{N}^2, \text{square-free numbers})$, and $(\{2^k\}, \text{odds})$ as the reader can easily check.

In Chapter 2 we present background on arithmetic functions. We recall the definitions of σ , ϕ , and μ and prove basic facts about them. Furthermore, we define convolution in the context of arithmetic functions to aid in our study of the underlying structure.

In Chapter 3 we give criteria for generalized arithmetic functions being multiplicative as well as presenting the basic information for factor sets and conjugate

pairs. We use the structure of generating functions and zeta functions to give a result on average orders of arithmetic functions of a certain form.

In Chapter 4 we study the convolution of σ_P and ϕ_Q for a conjugate pair (P, Q) . Indeed, we see that given conjugate pairs (P_1, Q_1) and (P_2, Q_2) , $\sigma_{P_1} * \phi_{Q_1} = \sigma_{P_2} * \phi_{Q_2}$. Thus to study the convolutions for general conjugate pairs, it suffices to study the simplest case: $P = \{1\}, Q = \mathbb{N}$ so that $\sigma_P = \phi_Q = I$. We find the average order of $\sigma * \phi$.

In Chapter 5 our focus is on products of σ_P and ϕ_Q for conjugate pairs (P, Q) . Our main result is Theorem 4.1.2, which states that given a conjugate pair (P, Q) we have

$$\frac{6}{\pi^2} = \frac{1}{\zeta(2)} < \frac{\sigma_P(n)\phi_Q(n)}{n^2} < \frac{\zeta(2)}{\zeta(4)} = \frac{15}{\pi^2}$$

which generalizes the well-known inequality

$$\frac{6}{\pi^2} < \frac{\sigma(n)\phi(n)}{n^2} < 1$$

In Chapter 6 we define inf- and sup-multipliers, and determine them for many σ_P and ϕ_Q . We also give results indicating that the primes in P and Q are the most important in finding the multipliers.

In Chapter 7 we look at compositions of σ_P and ϕ_Q when (P, Q) is a conjugate pair. We produce results regarding inequalities and average orders, as well as fixed points of the compositions. We further note that by taking sets P and Q which

are not conjugate, we may recover the well-known study of perfect numbers and introduce a new class of numbers - the pluperfects.

Chapter 2

Background

2.1 Basic Arithmetic Functions

Definition Let $f : \mathbb{N} \rightarrow \mathbb{N}$. Then we say that f is arithmetic.

Some examples of commonly studied arithmetic functions are σ , the sum-of-divisors function, ϕ , Euler's totient function, and the Möbius μ function. Properties of arithmetic functions that we are particularly interested in are multiplicativity and average orders. We define these now.

Definition Let f be an arithmetic function. Suppose that for any $r, s \in \mathbb{N}$ with $(r, s) = 1$ we have

$$f(rs) = f(r)f(s)$$

Then we say that f is multiplicative.

Definition Let f, g be two arithmetic functions. Then we say that f is of average order g , and write $f \sim g$, if

$$\lim_{N \rightarrow \infty} \frac{\sum_{n=1}^N f(n)}{\sum_{n=1}^N g(n)} = 1$$

Our main functions of interest, σ , ϕ , and μ , are multiplicative and have particularly simple average orders. We begin with a discussion of well-known results on σ , ϕ , and μ .

2.1.1 Sum-of-Divisors Function

The sum-of-divisors function σ is defined by

$$\sigma(n) = \sum_{d|n} d$$

Properties of the Sum-of-Divisors Function

As claimed above, σ is multiplicative (cf. [9]).

Proposition 2.1.1 *The sum-of-divisors function σ is multiplicative.*

Since σ is multiplicative, once we specify its value on all prime powers, we will have determined it completely. Given a prime p and $k \in \mathbb{N}$, note that

$$\sigma(p^k) = p^k + p^{k-1} + \dots + 1 = \frac{p^{k+1} - 1}{p - 1}$$

We can use this formula and the multiplicativity to evaluate σ for any $n \in \mathbb{N}$. Let $n = p_1^{e_1} \dots p_l^{e_l}$ where the p_i are distinct primes. Then

$$\sigma(n) = \sigma(p_1^{e_1} \dots p_l^{e_l}) = \prod_{i=1}^l \sigma(p_i^{e_i}) = \prod_{i=1}^l \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

In fact, σ is a special case of the function σ_x defined by

$$\sigma_x(n) = \sum_{d|n} d^x$$

Thus $d = \sigma_0$ gives the number of divisors of a given integer, and $\sigma = \sigma_1$. Furthermore, Proposition 2.1.1 can be easily generalized to see that σ_x is multiplicative for all $x \in \mathbb{R}_{\geq 0}$ and the corresponding evaluation formula as well. In particular note that

$$d(n) = d(p_1^{e_1} \dots p_l^{e_l}) = (e_1 + 1) \dots (e_l + 1)$$

Perfect Numbers

The earliest work on σ was on the so-called “perfect” numbers, defined by

Definition We say that $n \in \mathbb{N}$ is perfect if $\sigma(n) = 2n$.

Another way of saying this is that the sum of the proper divisors of n is equal to n :

Definition Let

$$s(n) = \sum_{\substack{d|n \\ d \neq n}} d$$

then we say that n is perfect if n is a fixed point of s , i.e. $s(n) = n$.

The main result in the study of perfect numbers is

Proposition 2.1.2 *Let $2^p - 1$ be a Mersenne prime. Then $n = 2^{p-1}(2^p - 1)$ is perfect. Furthermore, any even perfect number is of this form (i.e. given an even perfect number n , we get $n = 2^{p-1}(2^p - 1)$ for some Mersenne prime $2^p - 1$).*

Proof The first direction is fairly simple and is due to Euclid. The converse is somewhat more difficult, and was first proven by Euler.

Let $2^p - 1$ be a Mersenne prime, and set $n = 2^{p-1}(2^p - 1)$. Then

$$\begin{aligned} \sigma(n) &= \sigma(2^{p-1}(2^p - 1)) = \sigma(2^{p-1})\sigma(2^p - 1) \\ &= (2^p - 1)2^p = 2(2^{p-1}(2^p - 1)) = 2n \end{aligned}$$

Conversely, let n be an even perfect number. Then $n = 2^a b$ where $a > 0$ and b is odd. Thus

$$\sigma(n) = \sigma(2^a b) = \sigma(2^a)\sigma(b) = (2^{a+1} - 1)\sigma(b)$$

Then, since σ is perfect, we have

$$2^{a+1}b = (2^{a+1} - 1)\sigma(b)$$

or

$$\frac{b}{\sigma(b)} = \frac{2^{a+1} - 1}{2^{a+1}}$$

since the right-hand side of this equation is clearly in lowest terms, we know that

$$\begin{aligned} b &= (2^{a+1} - 1)c \\ \sigma(b) &= 2^{a+1}c \end{aligned}$$

for some $c \in \mathbb{N}$. Suppose $c > 1$, so that

$$\sigma(b) \geq b + c + 1 = 2^{a+1}c + 1 > 2^{a+1}c = \sigma(b)$$

a clear contradiction. Therefore $c = 1$, so $b = 2^{a+1} - 1$ and $\sigma(b) = 2^{a+1}$. Since $\sigma(b) = b + 1$ it is clear that b is prime. Thus $n = 2^a b$ where $b = 2^{a+1} - 1$ is a Mersenne prime. ■

This proposition establishes a one-to-one correspondence between even perfect numbers and Mersenne primes. It is not known at the time of this writing whether or not there are infinitely many such primes, and so whether or not there are

infinitely many even perfect numbers. Furthermore, there are no known odd perfect numbers though it has not been proven that they do not exist.

Behavior

Like many functions in number theory, σ behaves fairly erratically on a local level, but exhibits regularities when viewed globally. We have

$$n < \sigma(n) < n \left(e^{\gamma} \log \log n + \frac{.6483}{\log \log n} \right) \quad \text{for all } n > 2$$

with the lower bound coming directly from the definition and the upper bound from Robin [15]. Here γ is the Euler-Mascheroni constant defined by

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right) \approx .577$$

Since $\log \log n$ is very small, we see that $\sigma(n)$ is always very close to n . Furthermore, the following results show that these bounds cannot be improved significantly

$$\begin{aligned} \liminf \frac{\sigma(n)}{n} &= 1 \\ \limsup \frac{\sigma(n)}{n \log \log n} &= e^{\gamma} \end{aligned}$$

The first comes from the fact that $\sigma(p) = p + 1$ for all primes p , and the second is Gronwall's theorem [7]. On average we have

$$\sigma(n) \sim \frac{\pi^2}{6} n$$

cf Corollary 3.3.3.

Robin's inequality

As mentioned above, Guy Robin proved that

$$\sigma(n) < n \left(e^\gamma \log \log n + \frac{.6483}{\log \log n} \right)$$

He also conjectured the stronger

$$\sigma(n) < ne^\gamma \log \log n \quad \text{for all } n > 5040$$

and in fact proved that this statement is equivalent to the Riemann Hypothesis.

This is an indication that more than being an arithmetic curiosity, σ , and arithmetic functions in general, has deep connections to other parts of mathematics.

2.1.2 Euler's Totient Function

The totient function ϕ is defined by

$$\phi(n) = \#\{1 \leq k \leq n \mid \gcd(k, n) = 1\}$$

This function is also multiplicative (cf. [9]).

Proposition 2.1.3 *The totient function ϕ is multiplicative.*

Behavior

As with σ , we have bounds on ϕ as well as being able to compute its average order. In particular

$$\frac{n}{e^\gamma \log \log n + \frac{3}{\log \log n}} < \phi(n) < n \quad \text{for all } n > 2$$

The upper bound comes directly from the definition, while the lower bound can be found in e.g. [2]. Furthermore, as with σ , these bounds are very strong with respect to the actual behavior. We can see this by

$$\begin{aligned} \liminf \frac{\phi(n) \log \log n}{n} &= e^{-\gamma} \\ \limsup \frac{\phi(n)}{n} &= 1 \end{aligned}$$

The former equation relies on the prime number theorem, as seen in [9], while the latter comes from the fact that $\phi(p) = p - 1$ for all primes p .

Finally, the average order of ϕ is

$$\phi(n) \sim \frac{6}{\pi^2}n$$

cf Corollary 3.3.5.

2.1.3 The Möbius Function

Let $n = \prod_{i=1}^l p_i^{e_i}$, with the p_i distinct. Then Möbius's function is defined by

$$\mu(n) = \begin{cases} (-1)^l & e_i = 1 \text{ for all } i \\ 0 & \text{otherwise} \end{cases} \quad (2.1)$$

As claimed, μ is a multiplicative function, which is proven below.

Proposition 2.1.4 *The Möbius μ function is multiplicative.*

Proof Let $r, s \in \mathbb{N}$ with $\gcd(r, s) = 1$. If r or s is not square-free, say without loss of generality that r has a square factor, then rs will also not be square-free.

Thus

$$\mu(r)\mu(s) = 0\mu(s) = 0 = \mu(rs)$$

Otherwise, we can write $r = p_1 \dots p_l$ and $s = q_1 \dots q_m$ with all p_i, q_j distinct. Then

$$\mu(r)\mu(s) = (-1)^l(-1)^m = (-1)^{l+m} = \mu(rs)$$

thereby establishing multiplicativity. ■

The inequalities satisfied by μ are trivial and not of interest. However, we have that the average order of μ is 0

$$\mu(n) \sim 0$$

and in fact this statement is equivalent to the Prime Number Theorem (cf. [\[1\]](#)).

A related function to μ is the Mertens Function. Defined by

$$M(x) = \sum_{n \leq x} \mu(n)$$

Thus the statement of μ 's average order is equivalent to

$$\lim_{x \rightarrow \infty} \frac{M(x)}{x} = 0$$

Mertens ([\[12\]](#)) and Stieltjes ([\[19\]](#)) conjectured the much stronger statement

$$M(x) = O(x^{\frac{1}{2}})$$

which was disproven by Odlyzko and te Riele in 1985 ([13]). However, it is conjectured that the following slightly weaker growth rate holds:

$$M(x) = O(x^{\frac{1}{2}+\epsilon}) \quad \text{for all } \epsilon > 0$$

This has proven difficult to establish, however, and is equivalent to the Riemann Hypothesis.

We will see below μ 's crucial role in the study of arithmetic functions.

2.2 Convolution

While the above holds some interest, it may be lacking in structure. We attempt to remedy this by introducing the operation convolution. Using convolution we may proceed with a more systematic study of arithmetic, and in particular multiplicative, functions.

Definition Let f and g be two arithmetic functions. We define their convolution by

$$(f * g)(n) = \sum_{d_1 d_2 = n} f(d_1)g(d_2) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

Then we have the following:

Proposition 2.2.1 *The arithmetic functions $f : \mathbb{N} \rightarrow \mathbb{N}$ form a commutative unital ring under the operations function addition and convolution*

Proof This is mostly obvious or easy to check. We are, however, interested in the unit in the ring. We claim that

$$E(n) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases}$$

is the multiplicative identity. Observe that

$$(E * f)(n) = \sum_{d|n} E(d)f\left(\frac{n}{d}\right) = E(1)f(n) = f(n)$$

■

We now have a ring, which we call the Dirichlet ring, so we may ask about the unit group. Indeed it turns out that

Proposition 2.2.2 *Let f be an arithmetic function. Then f is a unit in the Dirichlet ring if and only if $f(1) \neq 0$.*

Proof First, let f be a unit in the Dirichlet ring. Then we know that there exists a g such that $f * g = E$. In particular, we have

$$1 = E(1) = (f * g)(1) = f(1)g(1)$$

Which is only possible if $g(1) = \frac{1}{f(1)}$ so that $f(1) \neq 0$.

Conversely, let f be given with $f(1) \neq 0$. Then as we saw, if f is to have an inverse, say g , we must have $g(1) = \frac{1}{f(1)}$. But then we see that

$$0 = E(2) = (f * g)(2) = f(1)g(2) + f(2)g(1)$$

and therefore

$$g(2) = -\frac{f(2)g(1)}{f(1)}$$

Continuing in this fashion we construct g while only dividing by $f(1)$. Thus as long as $f(1) \neq 0$ we may complete this construction so that f is a unit. ■

Recall that we are interested particularly in multiplicative functions, and in fact

Proposition 2.2.3 *The multiplicative arithmetic functions form a multiplicative subgroup of the unit group in the Dirichlet ring. In particular, multiplicativity is preserved under convolution.*

Proof That multiplicativity is preserved under convolution is the only non-obvious part of this statement. We therefore proceed to show this. Let f and g be multiplicative functions. Let $r, s \in \mathbb{N}$ with $(r, s) = 1$. Then

$$(f * g)(rs) = \sum_{d|rs} f(d)g\left(\frac{rs}{d}\right)$$

Now, because $(r, s) = 1$, we know that a divisor d of rs can be written uniquely as $d = d_1 d_2$ where $d_1 | r$, $d_2 | s$, and $(d_1, d_2) = 1$. Thus

$$\begin{aligned}
(f * g)(rs) &= \sum_{d|rs} f(d)g\left(\frac{rs}{d}\right) \\
&= \sum_{d_1|r} \sum_{d_2|s} f(d_1 d_2)g\left(\frac{rs}{d_1 d_2}\right) \\
&= \sum_{d_1|r} \sum_{d_2|s} f(d_1)f(d_2)g\left(\frac{r}{d_1}\right)g\left(\frac{s}{d_2}\right) \\
&= \sum_{d_1|r} f(d_1)g\left(\frac{r}{d_1}\right) \sum_{d_2|s} f(d_2)g\left(\frac{s}{d_2}\right) \\
&= (f * g)(r)(f * g)(s)
\end{aligned}$$

as desired. ■

In this context, we can clearly see the importance of μ from the following proposition:

Proposition 2.2.4 *Let $U(n) = 1$ for all n . Then $\mu = U^{-1}$, i.e. $\mu * U = E$.*

Proof First, since $\mu(1) = U(1) = 1$ we know that $(\mu * U)(1) = 1 = E(1)$. Thus we need only show that

$$(\mu * U)(n) = E(n) = 0 \quad \text{for all } n > 1$$

Let $n > 1$ be given. Then

$$(\mu * U)(n) = \sum_{d|n} \mu(d) U\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d)$$

Now, let $n = p_1^{e_1} \dots p_l^{e_l}$. Then

$$\begin{aligned} \sum_{d|n} \mu(d) &= \sum_{d|p_1^{e_1} \dots p_l^{e_l}} \mu(d) = \sum_{d|p_1 \dots p_l} \mu(d) \\ &= 1 + \sum_{i=1}^l \mu(p_i) + \sum_{i < j} \mu(p_i p_j) + \dots + \sum_{i_1 < \dots < i_l} \mu(p_{i_1} \dots p_{i_l}) \\ &= 1 - \binom{l}{1} + \binom{l}{2} - \dots + (-1)^l \binom{l}{l} \\ &= (1 - 1)^l = 0 \end{aligned}$$

■

We also get

Proposition 2.2.5 *Let $I(n) = n$ for all n . Then $\sigma = I * U$ and $\phi = I * \mu$.*

Proof The first is immediate from the definition:

$$(I * U)(n) = \sum_{d|n} I(d) U\left(\frac{n}{d}\right) = \sum_{d|n} d = \sigma(n)$$

To prove the second, we instead prove that $\phi * U = I$, and therefore $\phi = \phi * E = \phi * U * \mu = I * \mu$. Thus we wish to show that

$$\sum_{d|n} \phi(d) = n$$

Consider the set $\{\frac{1}{n}, \frac{2}{n}, \dots, \frac{n}{n}\}$. There are n elements in this set. Furthermore, we can write each as a reduced fraction $\frac{a}{d}$. Now, for each divisor d of n there will be exactly $\phi(d)$ fractions with d as denominator in the reduced fraction. Thus

$$\sum_{d|n} \phi(d) = n$$

as desired. ■

Note that this supplies proof of the multiplicativity of σ and ϕ in light of Proposition [2.2.3](#).

Chapter 3

Properties of Generalized Arithmetic Functions

3.1 Multiplicativity

As we stated above, we are often interested in arithmetic functions that are multiplicative. It is not surprising that the multiplicativity of μ_Q , U_P , ϕ_Q , and σ_P depend on the sets P and Q that are given. Recall the definition of a factor set (pg. 3). Then we have

Proposition 3.1.1 *μ_Q is multiplicative if and only if Q is a factor set.*

Proof Suppose that Q is a factor set, and let $r, s \in \mathbb{N}$ with $(r, s) = 1$. Then we get

$$\begin{aligned}\mu_Q(r)\mu_Q(s) &= \sum_{\substack{d|r \\ d \in Q}} \mu\left(\frac{r}{d}\right) \sum_{\substack{e|s \\ e \in Q}} \mu\left(\frac{s}{e}\right) = \sum_{\substack{d|r \\ d \in Q}} \sum_{\substack{e|s \\ e \in Q}} \mu\left(\frac{r}{d}\right) \mu\left(\frac{s}{e}\right) \\ &= \sum_{\substack{d|r \\ d \in Q}} \sum_{\substack{e|s \\ e \in Q}} \mu\left(\frac{rs}{de}\right) = \sum_{\substack{f|rs \\ f \in Q}} \mu\left(\frac{rs}{f}\right) = \mu_Q(rs)\end{aligned}$$

■

Proposition 3.1.2 U_P is multiplicative if and only if P is a factor set.

Proof Suppose that P is a factor set. Let $r, s \in \mathbb{N}$ with $(r, s) = 1$. Then if $rs \in P$ we must have $r, s \in P$ so that

$$U_P(rs) = 1 = U_P(r)U_P(s)$$

If $rs \notin P$ then either $r \notin P$ or $s \notin P$ so

$$U_P(rs) = 0 = U_P(r)U_P(s)$$

Conversely, suppose that U_P is multiplicative. Suppose that $rs \in P$ with $(r, s) =$

1. Then

$$1 = U_P(rs) = U_P(r)U_P(s)$$

so that $r, s \in P$. Further, if $r, s \in P$ with $(r, s) = 1$ then

$$U_P(rs) = U_P(r)U_P(s) = 1$$

so that $rs \in P$. ■

Furthermore, it turns out that ϕ_Q and σ_P have interpretations that appear to be direct generalizations of the classical σ and ϕ without utilizing μ_Q and U_P .

Proposition 3.1.3 *Given a factor set $Q \subset \mathbb{N}$ we have*

$$\phi_Q(n) = \#\{1 \leq k \leq n \mid (k, n) \in Q\}$$

Proof Fix a factor set $Q \subset \mathbb{N}$. By definition, we have $\phi_Q = I * \mu_Q$. Define

$$\hat{\phi}_Q(n) = \#\{1 \leq k \leq n \mid (k, n) \in Q\}$$

Then we wish to show that $\hat{\phi}_Q = \phi_Q$. We will first prove that these two functions agree on prime powers, and then that both are multiplicative.

We start by computing the value of ϕ_Q on prime powers. Let $p, n \in \mathbb{N}$ with p prime.

$$\phi_Q(p^n) = I * \mu_Q(p^n) = \sum_{d|p^n} I\left(\frac{p^n}{d}\right) \mu_Q(d) = \sum_{i=0}^n I(p^{n-i}) \mu_Q(p^i)$$

$$\begin{aligned}
&= \sum_{i=0}^n p^{n-i} \sum_{\substack{d \mid p^i \\ d \in Q}} \mu(p^i/d) = \sum_{i=0}^n p^{n-i} \sum_{j=0}^i \mu(p^{i-j}) \\
&= p^n + \sum_{i=1}^n p^{n-i} (\mu(1)U_P(p^i) + \mu(p)U_P(p^{i-1})) \\
&= p^n + \sum_{i=1}^n p^{n-i} (U_P(p^i) - U_P(p^{i-1}))
\end{aligned}$$

Now we move to $\hat{\phi}_Q$. Let $p, n \in \mathbb{N}$ with p prime.

$$\begin{aligned}
\hat{\phi}_Q(p^n) &= \#\{1 \leq k \leq p^n \mid (k, p^n) \in Q\} \\
&= \sum_{i=0}^n \#\{1 \leq k \leq p^n \mid (k, p^n) = p^i\} U_Q(p^i) \\
&= U_Q(p^n) + \sum_{i=0}^{n-1} (p^{n-i} - p^{n-i-1}) U_Q(p^i) \\
&= p^n + \sum_{i=1}^n p^{n-i} (U_Q(p^i) - U_Q(p^{i-1}))
\end{aligned}$$

which establishes agreement on prime powers.

We know that ϕ_Q is multiplicative, by Propositions 2.2.3 and 3.1.1. Thus, all that remains is to show that $\hat{\phi}_Q$ is multiplicative. Let $a, b \in \mathbb{N}$ with $(a, b) = 1$.

We wish to show that

$$\hat{\phi}_Q(a)\hat{\phi}_Q(b) = \hat{\phi}_Q(ab)$$

Consider the following isomorphism

$$f : \mathbb{Z}_{ab} \rightarrow \mathbb{Z}_a \times \mathbb{Z}_b$$

defined by

$$f(n) = (n_a, n_b)$$

where n_i is the reduction of $n \bmod i$. Now, let $S = \{1 \leq k \leq ab \mid (k, ab) \in Q\}$, and $T = \{1 \leq k \leq a \mid (k, a) \in Q\} \times \{1 \leq k \leq b \mid (k, b) \in Q\}$. We will show that $f(S) = T$.

Let $n \in S$, so $\gcd(n, ab) \in Q$. We have

$$\gcd(n, ab) = \gcd(n, a) \gcd(n, b) = \gcd(n_a, a) \gcd(n_b, b)$$

Then, since $\gcd(\gcd(n_a, a), \gcd(n_b, b)) = 1$ and Q is a factor set, we find that $\gcd(n_a, a) \in Q$ and $\gcd(n_b, b) \in Q$. Thus $f(n) = (n_a, n_b) \in T$. So we have $f(S) \subset T$.

Now let $(l, m) \in T$, so that $\gcd(l, a) \in P$ and $\gcd(m, b) \in Q$. By the Chinese Remainder Theorem, we can find a unique n with $1 \leq n \leq ab$ such that $l = n_a$ and $m = n_b$. Then

$$\gcd(n, ab) = \gcd(n, a) \gcd(n, b) = \gcd(n_a, a) \gcd(n_b, b) = \gcd(l, a) \gcd(m, b)$$

Since $\gcd(\gcd(l, a), \gcd(m, b)) = 1$ and Q is a factor set, we find that $\gcd(n, ab) \in Q$. Therefore $T \subset f(S)$. We have proven mutual containment, and therefore set equality. ■

Proposition 3.1.4 *Let $P \subset \mathbb{N}$. Then*

$$\sigma_P(n) = \sum_{\substack{d|n \\ d \in P}} \frac{n}{d}$$

Proof For σ_P it is easier since

$$\sigma_P(n) = I * U_P(n) = \sum_{d|n} I\left(\frac{n}{d}\right) U_P(d) = \sum_{\substack{d|n \\ d \in P}} \frac{n}{d} U_P(d) = \sum_{\substack{d|n \\ d \in P}} \frac{n}{d}$$

■

3.2 Conjugate Pairs

We recall the definition of a conjugate pair (P, Q) from page 3. Conjugate pairs are fundamental to the study of generalized functions, as exemplified by the following proposition seen in [3].

Proposition 3.2.1 *Suppose that (P, Q) is a conjugate pair. Then $U_P * \mu_Q = E$, or $U_P = \mu_Q^{-1}$.*

Proof We want to show

$$\sum_{d|n} \mu_Q\left(\frac{n}{d}\right) U_P(d) = E(n)$$

We use the following chain of equalities

$$\begin{aligned}
(U_P * \mu_Q)(n) &= \sum_{\substack{d|n \\ d \in P}} \mu_Q\left(\frac{n}{d}\right) = \sum_{\substack{de=n \\ d \in P}} \sum_{\substack{\delta D=e \\ \delta \in Q}} \mu(D) \\
&= \sum_{D|n} \sum_{\substack{d\delta=\frac{n}{D} \\ d \in P, \delta \in Q}} \mu(D) = \sum_{D|n} \mu(D) \sum_{\substack{d\delta=\frac{n}{D} \\ d \in P, \delta \in Q}} 1 \\
&= \sum_{D|n} \mu(D) = E(n)
\end{aligned}$$

■

3.3 Generating Functions

Given an arithmetic function f we define its Dirichlet generating series as follows

Definition Let f be an arithmetic function. Then its Dirichlet generating series is given by

$$DG(f; s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

Often, the Dirichlet generating series of a common arithmetic function will correspond to a recognizable zeta function. For example

$$DG(U; s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$$

So the Dirichlet generating series for U is the standard Riemann zeta function.

Furthermore, generating functions are compatible with convolution in the following sense.

Proposition 3.3.1 *Given two arithmetic functions f and g we have*

$$DG(f * g; s) = DG(f; s)DG(g; s)$$

Proof By definition we have

$$DG(f * g; s) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^s} = \sum_{n=1}^{\infty} \frac{\sum_{d_1 d_2 = n} f(d_1)g(d_2)}{n^s}$$

On the other hand, we know that

$$DG(f; s)DG(g; s) = \sum_{l=1}^{\infty} \frac{f(l)}{l^s} \sum_{m=1}^{\infty} \frac{g(m)}{m^s}$$

However, the denominator in this product will be n^s precisely when the numerator is $f(l)g(m)$ with $lm = n$. Thus the right-hand side of this last equation is the generating function of the convolution, as desired. ■

This proposition allows us to compute Dirichlet generating series that might not have been obvious to begin with. For example, it is not clear a priori what $DG(\sigma; s)$ will be. But, in light of this proposition, we get

$$DG(\sigma; s) = DG(I * U; s) = DG(I; s)DG(U; s) = \zeta(s-1)\zeta(s)$$

Similarly,

$$DG(\mu; s) = \frac{1}{DG(U; s)} = \frac{1}{\zeta(s)}$$

and

$$DG(\phi; s) = \frac{\zeta(s-1)}{\zeta(s)}$$

3.3.1 Average Order

Theorem 3.3.2 *Suppose $f : \mathbb{N} \rightarrow \mathbb{C}$ is $O(n^{1-\epsilon})$ for some $\epsilon > 0$. Then the Dirichlet convolution $I * f$ has average order $DG(f; 2)n$.*

Proof We observe that

$$\begin{aligned} \sum_{l=1}^N I * f(l) &= \sum_{n=1}^N \frac{\lfloor \frac{N}{n} \rfloor (\lfloor \frac{N}{n} \rfloor + 1)}{2} f(n) \\ &= \frac{1}{2} \sum_{n=1}^N \left(\frac{N}{n} + O(1) \right) \left(\frac{N}{n} + O(1) \right) f(n) \\ &= \frac{N^2}{2} \sum_{n=1}^N \frac{f(n)}{n^2} + O \left(N \sum_{n=1}^N \frac{f(n)}{n} \right) + O \left(\sum_{n=1}^N f(n) \right) \end{aligned}$$

Now, we know that

$$\begin{aligned} \sum_{n=N+1}^{\infty} \frac{f(n)}{n^2} &\leq \sum_{n=N+1}^{\infty} \frac{Cn^{1-\epsilon}}{n^2} = C \sum_{n=N+1}^{\infty} n^{-1-\epsilon} \\ &\leq C \int_N^{\infty} n^{-1-\epsilon} dn = DN^{-\epsilon} = O(N^{-\epsilon}) \end{aligned}$$

Similarly we have

$$\begin{aligned} \sum_{n=1}^N \frac{f(n)}{n} &\leq \sum_{n=1}^N \frac{Cn^{1-\epsilon}}{n} = C \sum_{n=1}^N n^{-\epsilon} \\ &\leq C \left(1 + \int_1^N n^{-\epsilon} dn \right) = O(N^{1-\epsilon}) \end{aligned}$$

and

$$\sum_{n=1}^N f(n) \leq \sum_{n=1}^N Cn^{1-\epsilon} \leq C \left(\int_1^N n^{1-\epsilon} dn \right) = O(N^{2-\epsilon})$$

Thus altogether we get

$$\sum_{l=1}^N I * f(l) = \frac{N^2}{2} \sum_{n=1}^{\infty} \frac{f(n)}{n^2} + O(N^{2-\epsilon})$$

Putting this into the average order notation yields

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{\sum_{l=1}^N I * f(l)}{\sum_{l=1}^N n DG(f; 2)} &= \lim_{N \rightarrow \infty} \frac{\frac{N^2}{2} \sum_{n=1}^{\infty} \frac{f(n)}{n^2} + O(N^{2-\epsilon})}{\frac{N^2+N}{2} DG(f; 2)} \\ &= \lim_{N \rightarrow \infty} \frac{\frac{N^2}{2} DG(f; 2) + O(N^{2-\epsilon})}{\frac{N^2}{2} DG(f; 2) + O(N)} = 1 \end{aligned}$$

Corollary 3.3.3 $\sigma \sim \frac{\pi^2}{6}n$

Proof We know that $\sigma = I * U$. Since $U = O(1) = O(n^{1-\epsilon})$ for $\epsilon = 1$ we have

$$\sigma \sim DG(U; 2)n = \zeta(2)n = \frac{\pi^2}{6}n$$

■

Corollary 3.3.4 $\sigma_{\{2^k\}} \sim \frac{4}{3}n$

Proof We know that $\sigma_{\{2^k\}} = I * U_{\{2^k\}}$. Since $U_{\{2^k\}} = O(1) = O(n^{1-\epsilon})$ for $\epsilon = 1$ we have

$$\begin{aligned} \sigma_{\{2^k\}} \sim DG(U_{\{2^k\}}; 2)n &= \sum_{m=1}^{\infty} \frac{U_{\{2^k\}}}{m^2} n = \sum_{k=1}^{\infty} \frac{1}{(2^k)^2} n \\ &= \sum_{k=1}^{\infty} \frac{1}{4^k} n = \frac{4}{3} n \end{aligned}$$

■

Corollary 3.3.5 $\phi \sim \frac{6}{\pi^2}n$

Proof Since $\phi = I * \mu$ and $\mu = O(1)$ we get

$$\phi \sim DG(\mu; 2)n = \frac{n}{\zeta(2)} = \frac{6}{\pi^2}n$$

■

Chapter 4

Products

4.1 Inequalities

We begin with σ and ϕ . It is well-known that

$$\frac{6}{\pi^2} < \frac{\sigma(n)\phi(n)}{n^2} < 1$$

for all $n > 1$. We will prove the former inequality in a more general statement later. For now we review the proof of the latter.

Proposition 4.1.1 *Let $n > 1$ be given. Then we have*

$$\frac{\sigma(n)\phi(n)}{n^2} < 1$$

Proof Let $n > 1$ be given, and suppose that $n = \prod_{i=1}^l p_i^{e_i}$ is its prime factorization. Then

$$\begin{aligned} \frac{\sigma(n)\phi(n)}{n^2} &= \prod_{i=1}^l \frac{\sigma(p_i^{e_i})\phi(p_i^{e_i})}{(p_i^{e_i})^2} = \prod_{i=1}^l \frac{\frac{p_i^{e_i+1}-1}{p_i-1}(p_i^{e_i}-p_i^{e_i-1})}{p_i^{2e_i}} \\ &= \prod_{i=1}^l \frac{p_i^{2e_i}-p_i^{e_i-1}}{p_i^{2e_i}} < \prod_{i=1}^l 1 = 1 \end{aligned}$$

■

To generalize the former inequality as well as proving a more general upper bound, we have

Theorem 4.1.2 *Let (P, Q) be a conjugate pair. Then*

$$\frac{1}{\zeta(2)} < \frac{\sigma_P(n)\phi_Q(n)}{n^2} < \frac{\zeta(2)}{\zeta(4)}$$

Proof We begin with the following observation: given a prime p , we have p in exactly one of P and Q , and p^2 in exactly one of P and Q . Thus we have the following four cases:

1. $p, p^2 \in P$
2. $p \in P, p^2 \in Q$
3. $p^2 \in P, p \in Q$
4. $p, p^2 \in Q$

Fix a prime p and an exponent $k \in \mathbb{N}$. Now, in the first case we have

$$\begin{aligned}\sigma_P\phi_Q(p^k) &< (p^k + p^{k-1} + \dots + 1)(p^k - p^{k-1} + p^{k-3}) \\ &< p^{2k} + p^{2k-3} + \dots + 1 < p^{2k} + p^{2k-2}\end{aligned}$$

and

$$\begin{aligned}\sigma_P\phi_Q(p^k) &> (p^k + p^{k-1} + p^{k-2})(p^k - p^{k-1}) \\ &= p^{2k} - p^{2k-3} > p^{2k} - p^{2k-2}\end{aligned}$$

In the second case, we get

$$\begin{aligned}\sigma_P\phi_Q(p^k) &< (p^k + p^{k-1} + p^{k-4} + \dots + 1)(p^k - p^{k-1} + p^{k-2}) \\ &< p^{2k} + p^{2k-3} + \dots + 1 < p^{2k} + p^{2k-2}\end{aligned}$$

and

$$\sigma_P\phi_Q(p^k) > (p^k + p^{k-1})(p^k - p^{k-1}) = p^{2k} - p^{2k-2}$$

In the third case

$$\begin{aligned}\sigma_P\phi_Q(p^k) &< (p^k + p^{k-2} + p^{k-4} + \dots + 1)(p^k - p^{k-2} + p^{k-3}) \\ &< p^{2k} + p^{2k-3} + \dots + 1 < p^{2k} + p^{2k-2}\end{aligned}$$

and

$$\sigma_P\phi_Q(p^k) > (p^k + p^{k-2})(p^k - p^{k-2}) = p^{2k} - p^{2k-4} > p^{2k} - p^{2k-2}$$

Finally, in the fourth case,

$$\begin{aligned}\sigma_P\phi_Q(p^k) &< (p^k + p^{k-3} + p^{k-4} \dots + 1)(p^k - p^{k-3} + p^{k-4}) \\ &= p^{2k} + p^{2k-3} + \dots + 1 < p^{2k} + p^{2k-2}\end{aligned}$$

and

$$\sigma_P\phi_Q(p^k) > (p^k)(p^k - p^{k-3}) = p^{2k} - p^{2k-3} > p^{2k} - p^{2k-2}$$

Thus, in any of the four cases, we have

$$p^{2k} - p^{2k-2} < \sigma_P\phi_Q(p^k) < p^{2k} + p^{2k-2}$$

or, dividing through by p^{2k} ,

$$1 - \frac{1}{p^2} < \frac{\sigma_P\phi_Q(p^k)}{p^{2k}} < 1 + \frac{1}{p^2}$$

Thus given $n \in \mathbb{N}$ we can write $n = p_1^{e_1} \dots p_l^{e_l}$ so that

$$\begin{aligned}\frac{1}{\zeta(2)} &< \prod_{i=1}^l 1 - \frac{1}{p_i^2} < \prod_{i=1}^l \frac{\sigma_P\phi_Q(p_i^{e_i})}{p_i^{2e_i}} \\ &< \prod_{i=1}^l 1 + \frac{1}{p_i^2} < \frac{\zeta(2)}{\zeta(4)}\end{aligned}$$

since

$$\prod_{i=1}^l \frac{\sigma_P \phi_Q(p_i^{e_i})}{p_i^{2e_i}} = \frac{\sigma_P \phi_Q(n)}{n^2}$$

we get the desired inequality. ■

Chapter 5

Growth Rates

Throughout this section, let $\mathcal{P}$ denote the set of all primes.

5.1 Definitions

We start with some definitions.

Definition Let $a : \mathbb{N} \rightarrow \mathbb{R}^+$ be a function. We say that a function f is a

1. sup multiplier for a if

$$0 < \limsup a(n)f(n) < \infty$$

2. inf multiplier for a if

$$0 < \liminf a(n)f(n) < \infty$$

3. normed sup multiplier for a if

$$\limsup a(n)f(n) = 1$$

4. normed inf multiplier for a if

$$\liminf a(n)f(n) = 1$$

5.2 Results

Recall from above that

$$\begin{aligned}\liminf \frac{\sigma(n)}{n} &= 1 \\ \limsup \frac{\phi(n)}{n} &= 1\end{aligned}$$

Or, in our new terminology, $\frac{1}{n}$ is a normed inf multiplier for $\sigma(n)$ and a normed sup multiplier for $\phi(n)$. In fact, these statements generalize cleanly to

Proposition 5.2.1 *$\frac{1}{n}$ is a normed inf multiplier for $\sigma_S(n)$ and a normed sup multiplier for $\phi_S(n)$ for all $S \subset \mathbb{N}$.*

Proof

$$\phi(n) \leq \phi_S(n) \leq n \leq \sigma_S(n) \leq \sigma(n)$$

so that

$$\frac{\phi(n)}{n} \leq \frac{\phi_S(n)}{n} \leq 1 \leq \frac{\sigma_S(n)}{n} \leq \frac{\sigma(n)}{n}$$

thus

$$1 = \limsup \frac{\phi(n)}{n} \leq \limsup \frac{\phi_S(n)}{n} \leq 1$$

giving equality throughout. Similarly

$$1 \leq \liminf \frac{\sigma_S(n)}{n} \leq \liminf \frac{\sigma(n)}{n} = 1$$

■

We also had that $\frac{1}{n \log \log n}$ is a sup multiplier for σ and $\frac{\log \log n}{n}$ is an inf multiplier for ϕ . We now generalize these results. We begin with a lemma.

Lemma 5.2.2 *Suppose that $S, T \subset \mathbb{N}$ such that $S \cap \mathcal{P} = T \cap \mathcal{P}$. Then*

1. *f is a sup multiplier for σ_S if and only if f is a sup multiplier for σ_T .*
2. *f is an inf multiplier for σ_S if and only if f is an inf multiplier for σ_T .*
3. *f is a sup multiplier for ϕ_S if and only if f is a sup multiplier for ϕ_T .*
4. *f is an inf multiplier for ϕ_S if and only if f is an inf multiplier for ϕ_T .*

Proof Let $p \in \mathcal{P} \cap S = \mathcal{P} \cap T$, and fix $k \in \mathbb{N}$. Then

$$1 - \frac{1}{p^2} \leq \frac{1 - \frac{1}{p^2}}{1 - \frac{1}{p^{k+1}}} = \frac{p^k + p^{k-1}}{p^k + p^{k-1} + \dots + 1}$$

$$\begin{aligned}
&\leq \frac{\sigma_S(p^k)}{\sigma_T(p^k)} \leq \frac{p^k + p^{k-1} + \dots + 1}{p^k + p^{k-1}} \\
&\leq \frac{1 - \frac{1}{p^{k+1}}}{1 - \frac{1}{p^2}} \leq \frac{1}{1 - \frac{1}{p^2}}
\end{aligned}$$

and if $p \notin \mathcal{P} \cap S = \mathcal{P} \cap T$, then

$$\begin{aligned}
1 - \frac{1}{p^2} &\leq 1 - \frac{\frac{1}{p^2}}{p-1} \leq 1 - \frac{\frac{1}{p^3} - \frac{1}{p^{k+1}}}{1 - \frac{1}{p}} \\
&\leq 1 - \frac{\frac{1}{p^3} - \frac{1}{p^{k+1}}}{1 - \frac{1}{p} + \frac{1}{p^2} - \frac{1}{p^{k+1}}} = \frac{p^k + p^{k-2}}{p^k + p^{k-2} + \dots + 1} \\
&\leq \frac{\sigma_S(p^k)}{\sigma_T(p^k)} \leq \frac{1}{1 - \frac{1}{p^2}}
\end{aligned}$$

Thus given any $n = p_1^{e_1} \dots p_l^{e_l}$ we will have

$$\begin{aligned}
\frac{6}{\pi^2} &= \prod_{p \in \mathcal{P}} 1 - \frac{1}{p^2} \leq \prod_{i=1}^l 1 - \frac{1}{p_i^2} \\
&\leq \frac{\sigma_S(n)}{\sigma_T(n)} \leq \prod_{i=1}^l \frac{1}{1 - \frac{1}{p_i^2}} \\
&\leq \prod_{p \in \mathcal{P}} \frac{1}{1 - \frac{1}{p^2}} = \frac{\pi^2}{6}
\end{aligned}$$

Similarly, let $p \in \mathcal{P} \cap S = \mathcal{P} \cap T$, $p > 2$. Then

$$\begin{aligned}
1 - \frac{1}{(p-1)^2} &\leq 1 - \frac{1}{p^2} = \frac{p^k - p^{k-2}}{p^k} \leq \frac{\phi_S(p^k)}{\phi_T(p^k)} \\
&\leq \frac{p^k}{p^k - p^{k-2}} = \frac{1}{1 - \frac{1}{p^2}} \leq \frac{1}{1 - \frac{1}{(p-1)^2}}
\end{aligned}$$

and for $p \notin \mathcal{P} \cap S = \mathcal{P} \cap T$, $p > 2$ we get

$$\begin{aligned}
1 - \frac{1}{(p-1)^2} &\leq 1 - \frac{p^{k-1} - 1}{(p-1)(p^k - p^{k-1} + p^{k-2} + \dots + 1)} \\
&\leq \frac{\phi_S(p^k)}{\phi_T(p^k)} \leq \frac{p^k - p^{k-1} + p^{k-2} + \dots + 1}{p^k - p^{k-1}} \\
&\leq \frac{1}{1 - \frac{1}{(p-1)^2}}
\end{aligned}$$

Let

$$L = \prod_{\substack{p \in \mathcal{P} \\ p > 2}} 1 - \frac{1}{(p-1)^2}$$

Then, given any $n = 2^e p_1^{e_1} \dots p_l^{e_l}$ we have

$$\begin{aligned}
\frac{L}{2} &= \frac{1}{2} \prod_{\substack{p \in \mathcal{P} \\ p > 2}} 1 - \frac{1}{(p-1)^2} \leq \frac{1}{2} \prod_{i=1}^l 1 - \frac{1}{(p_i-1)^2} \\
&\leq \frac{\phi_S(p_1^{e_1} \dots p_l^{e_l})}{2\phi_T(p_1^{e_1} \dots p_l^{e_l})} = \frac{2^{e-1} \phi_S(p_1^{e_1} \dots p_l^{e_l})}{2^e \phi_T(p_1^{e_1} \dots p_l^{e_l})} \\
&\leq \frac{\phi_S(n)}{\phi_T(n)} \leq \frac{2^e \phi_S(n)}{2^{e-1} \phi_T(n)} \\
&= \frac{2\phi_S(n)}{\phi_T(n)} \leq 2 \prod_{i=1}^l \frac{1}{1 - \frac{1}{(p_i-1)^2}} \\
&\leq 2 \prod_{\substack{p \in \mathcal{P} \\ p > 2}} \frac{1}{1 - \frac{1}{(p-1)^2}} = \frac{2}{L}
\end{aligned}$$

1. Suppose that f is a sup multiplier for σ_S . Then $\sigma_S f$ is bounded and has a subsequence not converging to 0. Therefore, $\sigma_T f = \sigma_T \frac{\sigma_S}{\sigma_T} f$ is bounded and has a subsequence not converging to 0 so that f is a sup multiplier for σ_T .

2. Suppose that f is an inf multiplier for σ_S . Then $\sigma_S f$ is bounded away from 0 and has a bounded subsequence. Therefore, $\sigma_T f = \sigma_T \frac{\sigma_S}{\sigma_T} f$ is bounded away from 0 and has a bounded subsequence so that f is an inf multiplier for σ_T .
3. Suppose that f is a sup multiplier for ϕ_S . Then $\phi_S f$ is bounded and has a subsequence not converging to 0. Therefore, $\phi_T f = \phi_T \frac{\phi_S}{\phi_T} f$ is bounded and has a subsequence not converging to 0 so that f is a sup multiplier for ϕ_T .
4. Suppose that f is an inf multiplier for ϕ_S . Then $\phi_S f$ is bounded away from 0 and has a bounded subsequence. Therefore, $\phi_T f = \phi_T \frac{\phi_S}{\phi_T} f$ is bounded away from 0 and has a bounded subsequence so that f is an inf multiplier for ϕ_T .

Theorem 5.2.3 *Let (P, Q) be a conjugate pair. Then the following are equivalent:*

1. $\sum_{p \in \mathcal{P} \cap P} \frac{1}{p}$ converges.
2. $f(n) = \frac{1}{n}$ is an inf multiplier for ϕ_Q .
3. $f(n) = \frac{1}{n}$ is a sup multiplier for σ_P .

Proof We begin by showing that (1) $\implies$ (2). Suppose that $\sum_{p \in \mathcal{P} \cap P} \frac{1}{p}$ converges. We need to show that $\frac{\phi_Q(n)}{n}$ is bounded away from 0 and has a bounded subsequence. Since $\sum_{p \in \mathcal{P} \cap P} \frac{1}{p}$ converges, we know that $\prod_{p \in \mathcal{P} \cap P} 1 - \frac{1}{p} = L > 0$.

Let $n = p_1^{e_1} \dots p_l^{e_l}$, with $p_1, \dots, p_r \in P$ and $p_{r+1}, \dots, p_l \in Q$. Then

$$\begin{aligned}
1 &\geq \frac{\phi_Q(n)}{n} = \prod_{i=1}^l \frac{\phi_Q(p_i^{e_i})}{p_i^{e_i}} \\
&= \prod_{i=1}^r \frac{\phi_Q(p_i^{e_i})}{p_i^{e_i}} \prod_{i=r+1}^l \frac{\phi_Q(p_i^{e_i})}{p_i^{e_i}} \\
&\geq \left[\prod_{i=1}^r 1 - \frac{1}{p_i} \right] \left[\prod_{i=r+1}^l 1 - \frac{1}{p_i^2} \right] \\
&\geq \frac{6}{\pi^2} \prod_{i=1}^r 1 - \frac{1}{p_i} \geq \frac{6}{\pi^2} L
\end{aligned}$$

We next need to show that (2) $\implies$ (3), i.e. that $\frac{\sigma_P(n)}{n}$ is bounded and has a subsequence bounded away from 0. In fact, by Theorem 4.1.2 and the above inequality we get

$$\frac{\sigma_P(n)}{n} < \frac{n\zeta(2)}{\phi_Q(n)\zeta(4)} \leq \frac{\pi^2}{6L}$$

while we know that $1 \leq \frac{\sigma_P(n)}{n}$. Thus $\frac{\sigma_P(n)}{n}$ is bounded from above and below away from 0.

Finally we have to show that (3) $\implies$ (1). Suppose that $\frac{1}{n}$ is a sup multiplier for σ_P . Let

$$n(x) = \prod_{\substack{p \leq x \\ p \in \mathcal{P} \cap P}} p$$

Then

$$\frac{\sigma_P(n(x))}{n(x)} = \prod_{\substack{p \leq x \\ p \in \mathcal{P} \cap P}} \frac{\sigma_P(p)}{p} = \prod_{\substack{p \leq x \\ p \in \mathcal{P} \cap P}} \frac{p+1}{p}$$

$$= \prod_{\substack{p \leq x \\ p \in \overline{\mathcal{P}} \cap P}} 1 + \frac{1}{p} = \prod_{\substack{p \leq x \\ p \in \overline{\mathcal{P}} \cap P}} \frac{1 - \frac{1}{p^2}}{1 - \frac{1}{p}}$$

Now, since $\frac{1}{n}$ is a sup multiplier for σ_P , we know that this is bounded from above and, since the product of the numerator converges, the product of the denominator must not go to 0. Therefore

$$\prod_{\substack{p \leq x \\ p \in \overline{\mathcal{P}} \cap P}} 1 - \frac{1}{p} \not\rightarrow 0$$

so that

$$\sum_{\substack{p \leq x \\ p \in \overline{\mathcal{P}} \cap P}} \frac{1}{p} \not\rightarrow \infty$$

and $\sum_{p \in \overline{\mathcal{P}} \cap P} \frac{1}{p}$ converges as desired. $\blacksquare$

Theorem 5.2.4 *Let (P, Q) be a conjugate pair, and suppose that P contains infinitely many primes. Then the following are equivalent*

1. $\frac{1}{n \log \log n}$ is a sup multiplier for σ_P .
2. $\frac{\log \log n}{n}$ is an inf multiplier for ϕ_Q .

Proof Suppose that (1) holds. Then

$$\limsup \frac{\sigma_P(n)}{n \log \log n} = \epsilon > 0$$

Therefore we can find a subsequence $\{n_j\}$ such that

$$\frac{\sigma_P(n_j)}{n_j \log \log n_j} \geq \frac{\epsilon}{2} \quad \text{for all } j$$

Now, from Theorem 4.1.2, this implies that

$$\frac{\phi_Q(n_j) \log \log n_j}{n_j} \leq \frac{\zeta(2)/\zeta(4)}{\epsilon/2} \quad \text{for all } j$$

which gives a bounded subsequence of $\frac{\phi_Q(n) \log \log n}{n}$, and therefore

$$\liminf \frac{\phi_Q(n) \log \log n}{n} < \infty$$

Furthermore

$$\liminf \frac{\phi_Q(n) \log \log n}{n} \geq \liminf \frac{\phi(n) \log \log n}{n} > 0$$

Thus establishing (2).

Conversely, suppose that (2) holds. So

$$\liminf \frac{\phi_Q(n) \log \log n}{n} = M < \infty$$

Then we have a subsequence $\{n_j\}$ such that

$$\frac{\phi_Q(n_j) \log \log n_j}{n_j} \leq 2M$$

Again, from Theorem 4.1.2, this implies that

$$\frac{\sigma_P(n_j)}{n_j \log \log n_j} \geq \frac{1}{2M\zeta(2)} \quad \text{for all } j$$

giving

$$\limsup \frac{\sigma_P(n)}{n \log \log n} > 0$$

Finally, recalling that

$$\limsup \frac{\sigma_P(n)}{n \log \log n} \leq \limsup \frac{\sigma(n)}{n \log \log n} < \infty$$

we establish (1). ■

Theorem 5.2.5 *Let (P, Q) be a conjugate pair, and suppose that Q contains finitely many primes. Then $\frac{1}{n \log \log n}$ is a sup multiplier for σ_P and $\frac{\log \log n}{n}$ is an inf multiplier for ϕ_Q .*

Proof Let (P, Q) be a conjugate pair, such that Q contains finitely many primes. Note that by Lemma 5.2.2, we may assume that $Q = \{p_1, \dots, p_l\}$ is a finite set of

primes. Define

$$L = \prod_{i=1}^l \frac{p_i + 1}{p_i}$$

We know that

$$\liminf \frac{\phi_Q(n) \log \log n}{n} \geq \liminf \frac{\phi(n) \log \log n}{n} > 0$$

Thus we need to show that

$$\liminf \frac{\phi_Q(n) \log \log n}{n} < \infty$$

Since $\liminf \frac{\phi(n) \log \log n}{n} < \infty$, we have a subsequence $\{n_j\}$ and $M \in \mathbb{N}$ such that

$$\frac{\phi(n_j) \log \log n_j}{n_j} < M \quad \text{for all } j$$

Now, let $a = p_1^{e_1} \dots p_l^{e_l} q_1^{f_1} \dots q_m^{f_m}$ be given. Then

$$\begin{aligned} \frac{\phi_Q(a)}{\phi(a)} &= \frac{\phi_Q(p_1^{e_1} \dots p_l^{e_l} q_1^{f_1} \dots q_m^{f_m})}{\phi(p_1^{e_1} \dots p_l^{e_l} q_1^{f_1} \dots q_m^{f_m})} \\ &= \prod_{i=1}^l \frac{\phi_Q(p_i^{e_i})}{\phi(p_i^{e_i})} \prod_{j=1}^m \frac{\phi_Q(q_j^{f_j})}{\phi(q_j^{f_j})} \\ &= \prod_{i=1}^l \frac{\phi_Q(p_i^{e_i})}{\phi(p_i^{e_i})} \leq \prod_{i=1}^l \frac{p_i^2 - 1}{p_i^2 - p_i} \\ &= \prod_{i=1}^l \frac{p_i + 1}{p_i} = L \end{aligned}$$

Thus

$$\begin{aligned}
\frac{\phi_Q(n_j) \log \log n_j}{n_j} &= \frac{\phi_Q(n_j)}{\phi(n_j)} \frac{\phi(n_j) \log \log n_j}{n_j} \\
&\leq L \frac{\phi(n_j) \log \log n_j}{n_j} \\
&< LM \quad \text{for all } j
\end{aligned}$$

Thus we have a bounded subsequence for $\frac{\phi_Q(n) \log \log n}{n}$ so

$$\liminf \frac{\phi_Q(n) \log \log n}{n} < \infty$$

and $\frac{\log \log n}{n}$ is an inf multiplier for ϕ_Q .

Then, by Theorem 5.2.4, we also have that $\frac{1}{n \log \log n}$ is a sup multiplier for σ_P .

■

Chapter 6

Convolutions

We now move to a study of $\sigma_P * \phi_Q$. It turns out that this function is the same for any conjugate pair (P, Q) , since

$$\sigma_P * \phi_Q = I * U_P * I * \mu_Q = I * I * U_P * \mu_Q = I * I$$

So their convolution is equal to the convolution of the identity function with itself no matter which conjugate pair is chosen. We note that

$$(I * I)(n) = \sum_{d|n} I(d)I\left(\frac{n}{d}\right) = \sum_{d|n} d \frac{n}{d} = \sum_{d|n} n = nd(n)$$

6.1 Average Order

The average order of $d(n)$ is $\log(n)$ (cf. [9]), therefore it is not surprising that

Theorem 6.1.1 *The function $\sigma_P * \phi_Q(n)$ has average order $n \log(n)$.*

Proof Since $\sigma_P * \phi_Q = I * I$ we will use this latter function. Note that

$$\sum_{n=1}^N I * I(n) = \frac{N^2}{2} \sum_{n=1}^N \frac{1}{n} + O(N^2) = \frac{N^2}{2} \log(N) + O(N^2)$$

Now,

$$\sum_{n=1}^N n \log(n) = \log\left(\prod_{n=1}^N n^n\right) = \log((N!)^N) - \sum_{n=1}^{N-1} \log(n!)$$

applying Stirling's formula, the above becomes

$$\begin{aligned} & N(N \log(N) - N + O(\log(N))) - \sum_{n=1}^{N-1} [n \log(n) - n + O(\log(n))] \\ &= N^2 \log(N) + O(N^2) - \sum_{n=1}^{N-1} n \log(n) = N^2 \log(N) + O(N^2) - \sum_{n=1}^N n \log(n) \end{aligned}$$

Thus

$$\sum_{n=1}^N n \log(n) = \frac{N^2}{2} \log(N) + O(N^2)$$

Putting these two results together we obtain

$$\lim_{N \rightarrow \infty} \frac{\sum_{n=1}^N I * I(n)}{\sum_{n=1}^N n \log(n)} = \lim_{N \rightarrow \infty} \frac{\frac{N^2}{2} \log(N) + O(N^2)}{\frac{N^2}{2} \log(N) + O(N^2)} = 1$$

establishing our claim that $\sigma_P * \phi_Q = I * I \sim N \log(N)$.

6.2 Limits

We have

Proposition 6.2.1 *The function $\frac{1}{2n}$ is a normed inf multiplier for $\sigma * \phi$.*

Proof We will show that there is a sequence $\{n_j\}$ with

$$\lim_{j \rightarrow \infty} \frac{\sigma * \phi(n_j)}{2n_j} = 1$$

and also that

$$\frac{\sigma * \phi(n)}{2n} \geq 1 \quad \text{for all } n > 1$$

For the first let p be prime. We have

$$\frac{\sigma * \phi(p)}{2p} = \frac{pd(p)}{2p} = 1$$

since there are infinitely many primes, this gives a subsequence with the desired property.

For the second, let $n > 1$ be given. Then

$$\frac{\sigma * \phi(n)}{2n} = \frac{nd(n)}{2n} = \frac{d(n)}{2} \geq 1$$

■

Chapter 7

Composition

7.1 Previous Results

We begin with a study of $\sigma \circ \phi$ without any associated subsets. We then proceed to generalizations.

7.1.1 Inequalities and Limits

Lower Limits

In 1964, Makowski and Schinzel conjectured that

$$\frac{\sigma(\phi(n))}{n} \geq \frac{1}{2}$$

for all n [11]. They were unable to prove this, however, and at the time even

$$\liminf \frac{\sigma(\phi(n))}{n} > 0$$

was unknown. This has since been established by Pomerance [14]. Kevin Ford provided the current best known explicit value [6]

$$\liminf \frac{\sigma(\phi(n))}{n} > \frac{1}{39.4}$$

Makowski and Schinzel were able to prove that

$$\liminf \frac{\sigma(\phi(n))}{n} \leq \frac{1}{2} + \frac{1}{2^{34} - 4}$$

Thus we have

$$\frac{1}{39.4} < \liminf \frac{\sigma(\phi(n))}{n} \leq \frac{1}{2} + \frac{1}{2^{34} - 4}$$

Upper Limits

Results on the $\limsup$ seem to come more easily, however. Makowski and Schinzel proved that

$$\limsup \frac{\sigma(\phi(n))}{n} = \infty$$

and Luca and Pomerance [10] proved the stronger

$$\limsup \frac{\sigma(\phi(n))}{n \log \log n} = e^\gamma$$

We will use their method to prove a more general statement about the lim sup for conjugate pairs later.

7.2 Conjugate Pairs

7.2.1 Inequalities and Limits

As stated above, for $(P, Q) = (\mathbb{N}, \{1\})$ we have

$$\limsup \frac{\sigma_P(\phi_Q(n))}{n \log \log n} = e^\gamma$$

We now generalize this statement.

Theorem 7.2.1 *Suppose that (P, Q) is a conjugate pair with $\mathcal{P} \subset P$. Define*

$$L_P = \limsup \frac{\sigma_P(n)}{n \log \log n}$$

Then

$$\limsup \frac{\sigma_P(\phi_Q(n))}{n \log \log n} = L_P$$

Proof We recall that

$$\liminf \frac{\phi(n) \log \log n}{n} = e^{-\gamma}$$

so that there is $c \in \mathbb{R}^+$ such that for sufficiently large n we get

$$\frac{cn}{\log \log n} < \phi(n) \leq \phi_Q(n) \leq n$$

Thus

$$\lim_{n \rightarrow \infty} \frac{\log \log \phi_Q(n)}{\log \log n} = 1$$

Now, note that

$$\begin{aligned} \limsup \frac{\sigma_P(\phi_Q(n))}{n \log \log n} &\leq \limsup \frac{\sigma_P(\phi_Q(n))}{\phi_Q(n) \log \log \phi_Q(n)} \\ &\leq \limsup \frac{\sigma_P(n)}{n \log \log n} \\ &= L_P \end{aligned}$$

For the reverse inequality, we let $\{n_k\}$ be a sequence such that

$$\lim \frac{\sigma(n_k)}{n_k \log \log n_k} = L_P$$

and, for each k , let p_k be the least prime with $p_k \equiv 1 \pmod{n_k}$. By Linnik's theorem, we know that there exists a constant S such that $n_k < p_k < n_k^S$, so that

$$\log n_k < \log p_k < S \log n_k$$

and

$$\log \log n_k < \log \log p_k < \log S + \log \log n_k$$

giving

$$\lim \frac{\log \log p_k}{\log \log n_k} = 1$$

Now, since Q contains no primes, we have

$$\begin{aligned} \limsup \frac{\sigma_P(\phi_Q(p_k))}{p_k \log \log p_k} &= \limsup \frac{\sigma_P(p_k - 1)}{p_k \log \log p_k} \\ &= \limsup \frac{(p_k - 1) \log \log n_k}{p_k \log \log p_k} \frac{\sigma_P(p_k - 1)}{(p_k - 1) \log \log n_k} \\ &\geq \limsup \frac{(p_k - 1) \log \log n_k}{p_k \log \log p_k} \frac{\sigma_P(n_k)}{(n_k) \log \log n_k} \\ &= \limsup \frac{\sigma_P(n_k)}{(n_k) \log \log n_k} \end{aligned}$$

which establishes our claim. $\blacksquare$

7.2.2 Fixed Points

We may then ask what can be said about the fixed points of compositions of σ_P and ϕ_Q when (P, Q) is a conjugate pair. Little has been established for general conjugate pairs, however we have studied some particular cases.

The Pair $P = \mathbb{N}, Q = \{1\}$

Here we obtain the classical arithmetic functions $\sigma_P = \sigma$ and $\phi_Q = \phi$. Many fixed points of $\sigma \circ \phi$ and $\phi \circ \sigma$ have been computed, but it seems that there has been little effort to classify them.

We first study fixed points of $\phi \circ \sigma$ and classify them according to their power of 2. We have established

Theorem 7.2.2 *Let n be a fixed point of $\phi \circ \sigma$ with $\text{ord}_2(n) < 4$. Then $n \in \{1, 2, 8, 12\}$.*

Proof We split the proof into cases based on $\text{ord}_2(n)$.

First, let $\text{ord}_2(n) = 0$. In other words, suppose that n is odd. Then $n = \phi(\sigma(n))$ implies that $\sigma(n) \in \{1, 2\}$ (since these are the only arguments where ϕ is odd). Thus $n = 1$.

Next, we let $\text{ord}_2(n) = 1$, so $n = 2a$ with a odd. Let $\sigma(a) = 3^k r$ with $(r, 3) = 1$. Then

$$2a = n = \phi(\sigma(2a)) = \phi(3^{k+1}r) = 2 * 3^k * \phi(r)$$

Thus

$$a = 3^k * \phi(r)$$

so $\phi(r) = 1$ and $a = 3^k$. Then $3 \nmid \sigma(a)$ so that $\sigma(a) = r \in \{1, 2\}$. Thus $a = r = 1$ and $n = 2$.

Now, let $\text{ord}_2(n) = 2$, so $n = 4a$ with a odd. Let $\sigma(a) = 7^k r$ with $(r, 7) = 1$.

Then

$$4a = n = \phi(\sigma(4a)) = \phi(7^{k+1}r) = 6 * 7^k * \phi(r)$$

so that

$$a = 3 * 7^k * \frac{\phi(r)}{2}$$

Since $4 \nmid \phi(r)$ we get three possibilities: $r = 4$, $r = p^\alpha$ or $r = 2p^\alpha$ for some odd prime p . If $r = 4$ then

$$a = 3 * 7^k$$

so that

$$\sigma(a) = 4 * \sigma(7^k)$$

and therefore $7 \nmid \sigma(a)$. Then $\sigma(a) = r = 4$ so $a = 3$ and $n = 12$, which is indeed a fixed point.

Otherwise, $r = p^\alpha$ or $r = 2p^\alpha$. Then $\phi(r) = p^{\alpha-1}(p-1)$. This gives

$$a = 3 * 7^k * p^{\alpha-1} * \frac{(p-1)}{2}$$

$$\sigma(a) \in \{7^k p^\alpha, 2 * 7^k p^\alpha\}$$

Let $b = \text{ord}_a(3)$. If b is odd then $4 \mid \sigma(3^b)$, which is impossible. Thus b is even. Furthermore, $p \equiv 3 \pmod{4}$, since $\frac{(p-1)}{2}$ is odd, and $p \equiv 1 \pmod{6}$, since b is even and therefore $3 \mid \frac{(p-1)}{2}$. These facts together tell us that $p \equiv 7 \pmod{12}$. Then, by Lemma 7.2.3, $p \nmid \sigma(3^b)$ and $7 \nmid \sigma(3^b)$ so that $\sigma(3^b) \nmid 2 * 7^k p^\alpha$, a clear contradiction.

Finally, let $n = 8a$ with a odd. Then let $\sigma(a) = 3^k 5^l r$ with $(3, r) = (5, r) = 1$. Then

$$n = 8a = \phi(\sigma(8a)) = \phi(15\sigma(a)) = \phi(3^{k+1} 5^{l+1} r) = 2 * 3^k * 4 * 5^l * \phi(r)$$

therefore

$$a = 3^k 5^l \phi(r)$$

so that $\phi(r)$ is odd. But then $r \in \{1, 2\}$, so $a = 3^k 5^l$ and $\sigma(a) \in \{3^k 5^l, 2 * 3^k 5^l\}$. Since $\sigma(a) > a$ if $a > 1$, we get that either $a = 1$ or $3^k 5^l$ is a perfect number. Obviously this latter is false, and therefore $a = 1$ so $n = 8$. ■

Lemma 7.2.3 *Let $p \equiv 7 \pmod{12}$ be prime, say $p = 12k + 7$. If $p \mid \sigma(3^n)$, then n is odd.*

Proof Let $p \equiv 7 \pmod{12}$ be prime with $p \mid \sigma(3^n)$. Then

$$\frac{3^{n+1} - 1}{2} = 3^n + \dots + 3 + 1 = cp$$

so

$$3^{n+1} \equiv 1 \pmod{p}$$

Let r be the order of 3 mod p . Then clearly $r \mid n+1$. Furthermore, by the law of quadratic reciprocity,

$$\left(\frac{3}{12k+7}\right) \left(\frac{12k+7}{3}\right) = (-1)^{\frac{(2)(12k+6)}{4}} = -1$$

Since $12k+7 \equiv 1 \pmod{3}$, this gives

$$\left(\frac{3}{12k+7}\right) = -1$$

in other words, 3 is not a square mod p . But this implies that 3 has even order mod p , since $\phi(p)$ is even, so r is even. Then $n+1$ is even, so n is odd as desired.

■

We may also examine the composition in the other order, that is $\sigma \circ \phi$. We have computed the fixed points of $\sigma \circ \phi$ up to 10^9 (cf. Table 7.1).

We also have the following theorem.

Theorem 7.2.4 *Let n be a square-free fixed point of $\sigma \circ \phi$ with $\omega(n) \leq 4$. Then $n \in \{3, 15, 255, 2418, 65535\}$.*

Proof We start with a definition

n	Prime Factorization of n	n	Prime Factorization of n
1	1	2835756	$2^2 * 3^3 * 7 * 11^2 * 31$
3	3	4059264	$2^7 * 3 * 11 * 31^2$
15	$3 * 5$	4451832	$2^3 * 3^2 * 7 * 11^2 * 73$
28	$2^2 * 7$	10890040	$2^3 * 5 * 7 * 19 * 23 * 89$
255	$3 * 5 * 17$	13192608	$2^5 * 3 * 11 * 13 * 31^2$
744	$2^3 * 3 * 31$	23001132	$2^2 * 3 * 7 * 11^2 * 31 * 73$
2418	$2 * 3 * 13 * 31$	54949482	$2 * 3^3 * 7^2 * 19 * 1093$
20440	$2^3 * 5 * 7 * 73$	110771178	$2 * 3 * 7 * 19 * 127 * 1093$
65535	$3 * 5 * 17 * 257$	220174080	$2^8 * 3 * 5 * 7 * 8191$
548856	$2^3 * 3^4 * 7 * 11^2$	445701354	$2 * 3 * 7^2 * 19 * 73 * 1093$

Table 7.1: All fixed points of $\sigma \circ \phi$ up to 1,000,000,000

Definition Let $n = p_1 \dots p_l$ be square-free. Then we say that n is distinguished if

$$\prod_{i \in I_1} p_i - 1 = \prod_{i \in I_2} p_i - 1 \implies I_1 = I_2$$

Otherwise we say that n is undistinguished.

we now present the proof of this theorem. We split the proof up into cases based on $\omega(n)$.

Prime Factors: 1

Suppose that p is prime and $p = \sigma(\varphi(p)) = \sigma(p - 1)$. Since $p - 1 + 1 = p$, we know that $p - 1$ is also prime. Therefore, $p = 3$ is the only prime fixed point.

Prime Factors: 2

We look at $n = pq$ with p, q distinct primes.

$$pq = \sigma(\varphi(pq)) = \sigma((p-1)(q-1))$$

It is clear that $1, p-1, q-1$, and $(p-1)(q-1)$ are all factors of this last, and they sum to pq . Thus, if these four factors are distinct, then there are no other divisors of $(p-1)(q-1)$. Then

$$d((p-1)(q-1)) = 4$$

Thus either

$$(p-1)(q-1) = ab$$

for primes a and b , which is impossible, or

$$(p-1)(q-1) = a^3$$

for some prime a . Then $p-1 = a$ and $q-1 = a^2$, so $q = (a-1)(a+1)$. Thus $a = 2$ and we get $p = 3$, $q = 5$, giving us the fixed point 15.

If these four factors are not distinct then, without loss of generality, $p = 2$ so that

$$2q = \sigma(q-1)$$

but then $q - 1 = p_1^{e_1}$, in which case p_1 divides the left-hand side of this equation but not the right, giving a contradiction.

Prime Factors: 3

Let p, q, r be distinct primes with $n = pqr$ a fixed point of $\sigma \circ \phi$. Suppose further that $p < q < r$.

n odd, distinguished

We have

$$\begin{aligned} pqr &= \sigma \circ \phi(pqr) \\ &= \sigma((p-1)(q-1)(r-1)) \end{aligned}$$

We are assuming that pqr is distinguished, and therefore $d((p-1)(q-1)(r-1)) = 8$. Thus we get that $(p-1)(q-1)(r-1)$ is of one of the forms a^7, ab^3, abc for primes a, b, c . Only the first is possible, and gives $p = 3, q = 5, r = 17$. Thus $n = 255$ is the only distinguished fixed point.

n even

Suppose that n is even and $n = pqr$. Without loss of generality, let $p = 2$.

$$\sigma(\phi(pqr)) = \sigma(\phi(2qr)) = \sigma((q-1)(r-1))$$

If we factor $(q-1)(r-1)$ as

$$(q-1)(r-1) = p_1^{e_1} p_2^{e_2} \dots p_l^{e_l}$$

then we get

$$2qr = \prod_{i=1}^l \sigma(p_i^{e_i})$$

This leaves the following possibilities

$$\begin{aligned} \frac{p_1^{e_1+1} - 1}{p_1 - 1} &= 2qr \\ \frac{p_1^{e_1+1} - 1}{p_1 - 1} &= 2q & \frac{p_2^{e_2+1} - 1}{p_2 - 1} &= r \\ \frac{p_1^{e_1+1} - 1}{p_1 - 1} &= 2, & \frac{p_2^{e_2+1} - 1}{p_2 - 1} &= qr \\ \frac{p_1^{e_1+1} - 1}{p_1 - 1} &= 2, & \frac{p_2^{e_2+1} - 1}{p_2 - 1} &= q, & \frac{p_3^{e_3+1} - 1}{p_3 - 1} &= r \end{aligned}$$

The third and fourth cases are impossible since $2 \notin \sigma(\mathbb{N})$. For case 1 we get

$$\begin{aligned} \frac{p_1^{e_1+1} - 1}{p_1 - 1} &= 2qr \\ p_1^{e_1} &= (q-1)(r-1) \end{aligned}$$

From the second equation we see that $p_1 = 2$, but then $\sigma(p_1^{e_1})$ would be odd, contradicting the first equation.

For case 2 we see

$$\frac{p_1^{e_1+1} - 1}{p_1 - 1} = 2q \quad \text{and} \quad \frac{p_2^{e_2+1} - 1}{p_2 - 1} = r$$

Since $\sigma(p_1^{e_1})$ is even, p_1 must be odd. Further,

$$(q - 1)(r - 1) = p_1^{e_1} p_2^{e_2}$$

is even and so $p_2 = 2$. Thus $2^{e_2+1} - 1 = r$. Substituting into the above we get

$$\begin{aligned} (q - 1)(2^{e_2+1} - 2) &= p_1^{e_1} 2^{e_2} \\ (q - 1) &= \frac{p_1^{e_1} 2^{e_2-1}}{2^{e_2} - 1} \end{aligned}$$

Note that

$$2q = p_1^{e_1} \dots + p_1 + 1$$

so $p_1 \nmid q - 1$. Then

$$p_1^{e_1} = 2^{e_2} - 1$$

We know using Catalan's conjecture that $e_1 = 1$. Therefore $p_1 = 2^{e_2} - 1$. Then

$$2q = \frac{p_1^{e_1+1} - 1}{p_1 - 1} = p_1 + 1 = 2^{e_2}$$

which is impossible.

n odd, undistinguished

The only way for n to be undistinguished is if $(p-1)(q-1) = (r-1)$. Then we get the following factors of $(p-1)(q-1)(r-1)$: $(r-1)^2, r-1, (p-1)(r-1), (q-1)(r-1), p-1, q-1, 1$ which sum to $r^2 + pr - qr + 3r + 1$.

We also get the factor, $\frac{(r-1)^2}{2}$, though this might not be distinct. The only way for it to be equal to one of the factors we already have is if $p = 3$ so that $r-1 = 2(q-1)$, in which case

$$\frac{(r-1)^2}{2} = (q-1)(r-1)$$

If $p = 3$ then

$$\sigma(4(q-1)^2) = \sigma(2(q-1)(r-1)) = 3qr = 3q(2q-1)$$

Suppose that $q \geq 11$. Then

$$\sigma(4(q-1)^2) \geq 7(q-1)^2 > 6q^2 - 3q = 3q(2q-1)$$

Thus $3 < q < 11$, so $q \in \{5, 7\}$. For $q = 5$ we get $r = 9$, and for $q = 7$ we get $r = 13$. Since 9 is not prime, the first is impossible, and we can check that $2*7*13$ is not a fixed point of $\sigma \circ \phi$. Thus $\frac{(r-1)^2}{2}$ is indeed a new factor.

We can easily check that

$$\frac{3}{2}(r-1)^2 + (p-1)(r-1) + (q-1)(r-1) + 1 > pqr$$

and therefore no such fixed point exists.

Prime Factors: 4

Suppose that we have $n = pqrs$ square-free with p, q, r, s prime. We start with n odd and distinguished.

n ***odd, distinguished***

In this case we get

$$pqrs = \sigma \circ \phi(pqrs) = \sigma((p-1)(q-1)(r-1)(s-1))$$

Since n is distinguished, we know that $d(\phi(pqrs)) = 16$. Thus, for primes a, b, c, d , we have one of the following:

$$(p-1)(q-1)(r-1)(s-1) = a^{15}$$

$$(p-1)(q-1)(r-1)(s-1) = a^7b$$

$$(p-1)(q-1)(r-1)(s-1) = a^3b^3$$

$$(p-1)(q-1)(r-1)(s-1) = a^3bc$$

$$(p-1)(q-1)(r-1)(s-1) = abcd$$

Since ord_2 of the left-hand side is at least 4, we are left with only the first two cases. In the second case, we must have $p-1, q-1, r-1, s-1 \in \{a, ab, a^2, a^3\}$. Thus $p-1 = a$, so that $p = 3$ and $a = 2$. But then $q-1, r-1$, or $s-1$ must be a^3 , but 9 is not prime. This eliminates the second case. Therefore $(p-1)(q-1)(r-1)(s-1) = a^{15}$. Thus each of p, q, r, s is of the form $a^k + 1$ and therefore a is even. Since a is also prime, we know that $a = 2$. Then $p-1 = 2$, $q-1 = 4$, $r-1 = 16$, $s-1 = 256$ giving $n = 3 * 5 * 17 * 257 = 65535$, which is a fixed point.

n even

Suppose that $n = 2qrs$, with q, r, s odd, distinct primes. We have $\sigma(\varphi(2qrs)) = 2qrs$. Let $\varphi(2qrs) = \prod_{i=1}^l p_i^{e_i}$. Thus $2qrs = \prod_{i=1}^l \sigma(p_i^{e_i})$. Since $2 \notin \sigma(\mathbb{N})$, we get the following possibilities:

$$2qrs = \sigma(p_1^{e_1})$$

$$2qr = \sigma(p_1^{e_1}), \quad s = \sigma(p_2^{e_2})$$

$$2q = \sigma(p_1^{e_1}), \quad rs = \sigma(p_2^{e_2})$$

$$2q = \sigma(p_1^{e_1}), \quad r = \sigma(p_2^{e_2}), \quad s = \sigma(p_3^{e_3})$$

1. Suppose that $l = 1$ and $2qrs = \sigma(p_1^{e_1})$.

Then p_1 must odd, for the output to be even. Recall that we also have

$(q-1)(r-1)(s-1) = p_1^{e_1}$. Then $2^3 \mid (q-1)(r-1)(s-1)$, but $2^3 \nmid p_1^{e_1}$, a contradiction.

2. Suppose that $l = 2$ and $2qr = \sigma(p_1^{e_1}), s = \sigma(p_2^{e_2})$.

Recall that we have $(q-1)(r-1)(s-1) = p_1^{e_1}p_2^{e_2}$. Then $2^3 \mid (q-1)(r-1)(s-1)$ so $e_2 \geq 3$ and p_1 is odd. It follows that $p_2 = 2$ and $s = 2^{e_2+1} - 1$. Then we know that $2^{e_2} - 1 = p_1^a$ with $a \leq e_1$. By Catalan's conjecture we know that both e_2 and $e_2 + 1$ are prime. So $e_2 = 2$, which is impossible since $e_2 \geq 3$.

3. Suppose that $l = 2$ and $2q = \sigma(p_1^{e_1}), rs = \sigma(p_2^{e_2})$.

We find that $p_2 = 2$ and p_1 is an odd prime. Thus e_1 is odd, so $e_1 + 1$ is even. We say $e_1 + 1 = 2k$. Then

$$2q = \frac{p_1^{e_1+1}}{p_1 - 1} = \frac{p_1^{2k} - 1}{p_1 - 1} = \frac{(p_1^k - 1)(p_1^k + 1)}{p_1 - 1}$$

In order for this last term to be twice a prime, we must have $k = 1$. Therefore $e_1 = 1$, and $2q = p_1 + 1$. Furthermore, $p_1 \nmid q - 1$ so that $q - 1 = 2^a$. Thus $q = 2^a + 1$ is prime, as is $2q - 1 = 2^{a+1} + 1$. Therefore $a = 1$, $q = 3$, and $p_1 = 5$.

Since $e_1 = 1$ we have precisely one of $5 \mid r - 1$ and $5 \mid s - 1$. Say without loss of generality that $r = 2^b + 1$ and $s = 5 * 2^c + 1$. This gives us

$$5 * 2^{b+c} + 5 * 2^c + 2^b + 1 = (2^b + 1)(5 * 2^c + 1) = 2^{e_2+1} - 1$$

Then since $p = 2$ and $q = 3$, we must have $r \geq 5$ so $b \geq 2$. Therefore looking at this equation mod 4 we get $c = 1$, so $s = 11$. Now we have

$$11(2^b + 1) = 2^{e_2+1} - 1$$

We know that $e_2 \geq 3$, so looking mod 8 gives

$$3 * 2^b \equiv 4 \pmod{8}$$

so that $b \leq 2$. We saw above that $b \geq 2$, so $b = 2$. Thus $r = 5$. But then $rs = 55 \neq 2^{e_2+1} - 1$. Thus there are no fixed points of this form.

4. Suppose that $l = 3$ and $2q = \sigma(p_1^{e_1}), r = \sigma(p_2^{e_2}), s = \sigma(p_3^{e_3})$. We know that

$$p_1^{e_1} p_2^{e_2} p_3^{e_3} = \varphi(2qrs) = (q-1)(r-1)(s-1)$$

and therefore $p_i = 2$ for some i . Since $2q = \sigma(p_1^{e_1})$ is even we know that $p_1 \neq 2$. Assume, without loss of generality, that $p_3 = 2$, and note that $e_3 \geq 3$. Thus $s = \sigma(2^{e_3}) = 2^{e_3+1} - 1$. We know have that $s-1 \mid p_1^{e_1} p_2^{e_2} 2^{e_3}$ and since $4 \nmid s-1$, $\frac{s-1}{2} \mid p_1^{e_1} p_2^{e_2}$.

Now, $\frac{s-1}{2} = 2^{e_3} - 1$, and since s is prime, we know that $e_3 + 1$ is a prime larger than 3, say $e_3 + 1 = p = 2k + 1$, $k \geq 2$. Thus $\frac{s-1}{2} = 2^{2k} - 1 = (2^k - 1)(2^k + 1)$, and therefore $(2^k - 1)(2^k + 1) \mid p_1^{e_1} p_2^{e_2}$. Since $\gcd(2^k - 1, 2^k + 1) = 1$, we get

$2^k - 1$ and $2^k + 1$ are powers of either p_1 or p_2 . By Catalan's conjecture, then, $k = 2$ or $k = 3$. Thus p_1 and p_2 are either 3 and 5, or 3 and 7. Recall that $\sigma(p_1^{e_1}) = 2q$, with q odd, so that $\sigma(p_1^{e_1}) \equiv 2 \pmod{4}$. Then $p_1 \equiv 1 \pmod{4}$ so $p_1 = 5$ and $p_2 = 3$. Furthermore, $k = 2$ and $s = 31$.

We have established that $p_1 = 5$, and since $\sigma(p_1^{e_1}) \equiv 0 \pmod{2}$ we know that e_1 is odd. We get $\frac{5^{e_1+1}-1}{4} = 2q$ so that $5^{e_1+1} - 1 = 8q$. Viewing this equation mod 6 we get $(-1)^{e_1+1} - 1 \equiv 2q \pmod{6}$. Recalling that e_1 is odd, this means that $2q \equiv 0 \pmod{6}$. Thus $q \equiv 0 \pmod{3}$ and since q is prime we get $q = 3$, and therefore $e_1 = 1$.

Putting together our results so far we get

$$5 * 3^{e_2} * 2^4 = 2 * (r - 1) * 30$$

$$3^{e_2-1} * 2^2 = r - 1$$

We also know that $r = \frac{3^{e_2+1}-1}{2}$ so $r - 1 = \frac{3^{e_2+1}-3}{2}$. Thus $3 \mid r - 1$ but $9 \nmid r - 1$. Therefore $e_2 = 2$ and $r = 13$.

Thus $n = 2 * q * r * s = 2 * 3 * 13 * 31 = 2418$ is a fixed point of this form and there are no others.

n odd, undistinguished

For n undistinguished and odd, there are three possibilities:

$$\begin{aligned}(p-1)(q-1) &= r-1 \\ (p-1)(q-1)(r-1) &= s-1 \\ (p-1)(q-1) &= (r-1)(s-1)\end{aligned}$$

Suppose first that $(p-1)(q-1) = r-1$. Then $(r-1)(s-1)$ and $r-1$ are repeated factors. Thus the sum of the known factors is $pqr s - s(r-1)$. We know that $\frac{(r-1)^2(s-1)}{2}$ is another factor, so the question is whether or not it is a new factor. Equating it with each of the currently known factors, we find that it is a new one if and only if $p, q, r, s \neq 3$. It is clear that $r \neq 3$, and without loss of generality we may assume that $q \neq 3$. So we are left with the possibilities $p = 3$ and $s = 3$.

Suppose first that $p = 3$, then $2(q-1) = (r-1)$, so $r = 2q-1$. Then look at the factor $\frac{(q-1)(r-1)(s-1)}{2}$. If this is a new factor, then

$$\begin{aligned}\sigma(\phi(pqr s)) &\geq pqr s - s(r-1) + \frac{(q-1)(r-1)(s-1)}{2} \\ &= pqr s + (q-1)(q-3) \left(s - \frac{q-1}{q-3} \right) > pqr s\end{aligned}$$

Which would contradict $pqr s$ being a fixed point. Thus it is a repeated factor, and therefore $s = 5$. So, we have $p = 3$, $r = 2q-1$, $s = 5$ and $pqr s$ a fixed point.

This gives

$$\begin{aligned}
15q(2q-1) &= pqr s = \sigma(\phi(pqr s)) \\
&= \sigma((p-1)(q-1)(r-1)(s-1)) = \sigma(16(q-1)^2) \\
&\geq 31(q-1)^2 = 31q^2 - 62q + 31
\end{aligned}$$

Then

$$30q^2 - 15q = 15q(2q-1) \geq 31q^2 - 62q + 31$$

so that $0 \geq q^2 - 47q + 31$ and therefore $q \leq 46$. By examination, we have no fixed points of the form $15q(2q-1)$ with $q \leq 46$. This eliminates the case where $p = 3$.

Suppose then that $s = 3$. Note that $4(r-1)$ is a factor of $(p-1)(q-1)(r-1)(s-1)$. If it is a new factor, then $\sigma(\phi(pqr s)) \geq pqr s - s(r-1) + 4(r-1) > pqr s$, which would be a contradiction. Thus it must be a repeated factor. Equating it to each of the current factors, we see that it is new unless p or q is 5. Without loss of generality, suppose that $p = 5$. Then we have $p = 5, r = 2q-1, s = 3$, giving

$$\begin{aligned}
15q(2q-1) &= pqr s = \sigma(\phi(pqr s)) \\
&= \sigma((p-1)(q-1)(r-1)(s-1)) = \sigma(32(q-1)^2) \\
&\geq 63(q-1)^2 = 63q^2 - 126q + 63
\end{aligned}$$

Then

$$30q^2 - 15q = 15q(2q - 1) \geq 63q^2 - 126q + 63$$

so that $0 \geq 33q^2 - 111q + 63$ and therefore $q < 3$, a clear contradiction.

Therefore, the cases where $\frac{(r-1)^2(s-1)}{2}$ is a repeated factor are impossible so that it is a new factor. But then

$$pqrs = \sigma(\phi(pqrs)) \geq pqrs - s(r-1) + \frac{(r-1)^2(s-1)}{2} > pqrs$$

So we get no fixed points of this form.

Suppose now that we are in the case where $(p-1)(q-1)(r-1) = s-1$. The sum of the known factors is now $pqrs - (s-1)$. Look at the factor $\frac{(s-1)^2}{2}$. It is distinct, unless p, q , or r is 3. Without loss of generality, let $p = 3$.

Then we have

$$\begin{aligned} 3qrs &= \sigma(4(q-1)^2(r-1)^2) \\ &\geq 7(q-1)^2(r-1)^2 + 3(q-1)^2(r-1) \end{aligned}$$

Since $s = 2(q-1)(r-1) + 1$ we get

$$0 \geq q^2r^2 - 5q^2r + 4q^2 - 8qr^2 + 13qr - 8q + 7r^2 - 11r + 4$$

Then we get $8 \leq q \leq 10$ and $5 \leq r \leq 31$. By inspection, there are no fixed points with these parameters. Thus $\frac{(s-1)^2}{2}$ is a new factor, so that

$$\sigma(\phi(pqrs)) \geq pqrs - (s-1) + \frac{(s-1)^2}{2} > pqrs$$

So there are no fixed points of this form.

Finally, suppose that $(p-1)(q-1) = (r-1)(s-1)$. Then the known factors sum up to $pqrs - (r-1)(s-1)$, because $(r-1)(s-1)$ was counted twice. We examine the factor $\frac{(r-1)^2(s-1)^2}{2}$. It must be repeated, and therefore, without loss of generality, $p = 3$. Then we have $2(q-1) = (r-1)(s-1)$. This gives

$$\begin{aligned} 3 \left(\frac{(r-1)(s-1)}{2} + 1 \right) rs &= pqrs \\ &= \sigma(4(q-1)^2) \\ &\geq 7(q-1)^2 \\ &= \frac{7(r-1)^2(s-1)^2}{4} \end{aligned}$$

There are only finitely many integer solutions to this inequality, and by examination, none of them yield a fixed point. This exhausts all the undistinguished possibilities, giving no undistinguished fixed points. ■

The Pair $P = \{n \mid n \text{ is square-free}\}, Q = \mathbb{N}^2$

Theorem 7.2.5 *Let $P = \{n \mid n \text{ is square-free}\}, Q = \mathbb{N}^2$. Then $\sigma_P(\phi_Q(p^k)) = p^k$ for a prime p and $k \in \mathbb{N}$ precisely when $p^k \in \{3, 4, 32\}$.*

Proof Suppose that $\sigma_P(\phi_Q(p^k)) = p^k$. Let $\phi_Q(p^k) = \prod_{i=1}^l q_i^{a_i}$. Then

$$p^k = \sigma_P(\phi_Q(p^k)) = \sigma_P\left(\prod_{i=1}^l q_i^{a_i}\right) = \prod_{i=1}^l (q_i^{a_i} + q_i^{a_i-1} + \dots + 1)$$

Thus for each i , either $q_i = p$ or $a_i = 1$. However, we know that $\phi_Q(p^k) = p^k - p^{k-1} + \dots + (-1)^k$ so that $p \nmid \phi_Q(p^k)$. Therefore, $\phi_Q(p^k) = q_1 \dots q_l$ is a product of distinct primes. Furthermore, for each i we have $q_i + 1 = p^{r_i}$, so $p^{r_i} - 1$ is prime. Thus either $p = 2$ or $r_i = 1$ for all i . If $r_i = 1$ then $q_i + 1 = p$ with both q_i and p prime so that $p = 3$ and $q_i = 2$. So we have either $p^k = 3$ or $p = 2$.

Then to continue, suppose that

$$2^k = \sigma_P(\phi_Q(2^k))$$

As established above, $\phi_Q(2^k) = q_1 \dots q_l$, a product of distinct primes. Therefore

$$2^k = \sigma_P(q_1 \dots q_l) = (q_1 + 1) \dots (q_l + 1)$$

So $q_i = 2^{e_i} - 1$ for each i . This gives

$$\prod_{i=1}^l (2^{e_i} - 1) = \prod_{i=1}^l q_i = \phi_Q(2^k) = \frac{2^{k+1} + (-1)^k}{3}$$

Therefore

$$3 \prod_{i=1}^l (2^{e_i} - 1) = 2^{k+1} + (-1)^k$$

Without loss of generality, we may assume that $1 < e_1 < \dots < e_l$. Now, look at the above equation mod 2^{e_1} . Then

$$3 \prod_{i=1}^l (-1) \equiv (-1)^k \pmod{2^{e_1}}$$

So

$$3 \equiv \pm 1 \pmod{2^{e_1}}$$

and therefore $e_1 = 2$. So we have either $l = 1$ and $2^k = \sigma_P(3) = 4$ or

$$9 \prod_{i=2}^l (2^{e_i} - 1) = 2^{k+1} + (-1)^k$$

Looking at this equation mod (2^{e_2}) gives

$$3 * 3 \equiv \pm 1 \pmod{2^{e_2}}$$

so that $e_2 = 3$ ($e_2 = 2$ is also a solution, but we know that $e_2 > e_1$). Then either $l = 2$ and $2^k = \sigma_P(3 * 7) = 32$ or

$$3 * 3 * 7 \prod_{i=3}^l (2^{e_i} - 1) = 2^{k+1} + (-1)^k$$

Again looking mod 2^{e_3} we have

$$3 * 3 * 7 \equiv \pm 1 \pmod{2^{e_3}}$$

so $e_3 \in \{2, 3, 4, 5, 6\}$. However, $e_3 \neq 4, 6$ since $2^{e_3} - 1$ must be prime and $e_3 > e_2 = 3$. This leaves only $e_3 = 5$. Then if $l = 3$ we would have $2^k = \sigma_P(3 * 7 * 31) = 1024$, however $3 * 7 * 31 \neq \phi_Q(1024) = 683$. Thus we must have $l > 3$ and

$$3 * 3 * 7 * 31 \prod_{i=4}^l (2^{e_i} - 1) = 2^{k+1} + (-1)^k$$

Take this equation mod 2^{e_4} to get

$$3 * 3 * 7 * 31 \equiv \pm 1 \pmod{2^{e_4}}$$

The solutions to this equation are $e_4 = \{2, 3, 4, 5\}$, but $e_4 > e_3 = 5$, a contradiction. Therefore, the only fixed points that we found are $2^k = 4, 32$. Combining this with our first conclusion that either $p^k = 3$ or $p = 2$ we get that the fixed points are precisely $\{3, 4, 32\}$ as claimed. ■

7.3 Non-Conjugate Pairs

7.3.1 Perfect Numbers

We have stated above the definition of a perfect number. That is, a number n such that $\sigma(n) = 2n$. This can in fact be phrased in the language of fixed points of compositions of generalized arithmetic functions.

Theorem 7.3.1 *Let $P = \mathbb{N}$ and $Q = \{\text{odd numbers}\}$. Then $n > 1$ is perfect if and only if it is a fixed point of $\phi_Q \circ \sigma_P$.*

Proof We start by noting that $\sigma_P = \sigma$ and giving a formula for ϕ_Q . It is easy to see that

$$\phi_Q(n) = \begin{cases} n & n \text{ odd} \\ \frac{n}{2} & n \text{ even} \end{cases}$$

Now, suppose that n is a perfect number. Then

$$\phi_Q(\sigma(n)) = \phi_Q(2n) = n$$

Conversely, suppose that

$$\phi_Q(\sigma(n)) = n$$

Now, if $\sigma(n)$ were odd, then

$$n = \phi_Q(\sigma(n)) = \sigma(n) > n$$

a clear contradiction. So $\sigma(n)$ is even. Then

$$n = \phi_Q(\sigma(n)) = \frac{\sigma(n)}{2}$$

and therefore

$$2n = \sigma(n)$$

as desired ■

7.3.2 Pluperfect Numbers

We have also studied the map $\phi_Q \circ \sigma_P$ for the pair $P = \mathbb{N}$, $Q = \mathbb{N}^2$. We call the fixed points pluperfect numbers (see Table 7.2 for all pluperfect numbers up to 10^8). Pluperfect numbers come in three types: basic, regular, and irregular. We call a pluperfect number n basic if it is of the form

$$n = 2^k, k \geq 1$$

We call a pluperfect number n regular if it is of the form

$$n = 2^k \prod_{i=1}^l p_i, \quad k \in \{1, 3, 7, 15, 31\}, l \geq 1$$

And we call a pluperfect number n irregular otherwise.

We have the following conjecture:

n	Prime Factorization of n	n	Prime Factorization of n
1		3591	$3^3 * 7 * 19$
2	2	4092	$2^2 * 3 * 11 * 31$
3	3	7182	$2 * 3^3 * 7 * 19$
6	$2*3$	11988	$2^2 * 3^4 * 37$
8	2^3	12040	$2^3 * 5 * 7 * 43$
21	$3*7$	32768	2^{15}
24	$2^3 * 3$	98304	$2^{15} * 3$
42	$2*3*7$	192640	$2^7 * 5 * 7 * 43$
60	$2^2 * 3 * 5$	382270	$2 * 5 * 7 * 43 * 127$
128	2^7	688128	$2^{15} * 3 * 7$
132	$2^2 * 3 * 11$	1486080	$2^8 * 3^3 * 5 * 43$
168	$2^3 * 3 * 7$	1529080	$2^3 * 5 * 7 * 43 * 127$
384	$2^7 * 3$	13037794	$2 * 7 * 11 * 31 * 2731$
720	$2^4 * 3^2 * 5$	24465280	$2^7 * 5 * 7 * 43 * 127$
1440	$2^5 * 3^2 * 5$	29719872	$2^6 * 3^6 * 7^2 * 13$
2688	$2^7 * 3 * 7$	49315840	$2^{15} * 5 * 7 * 43$
3010	$2*5*7*43$	52151176	$2^3 * 7 * 11 * 31 * 2731$

Table 7.2: All pluperfect numbers up to 100,000,000

Conjecture 7.3.2 *Regular pluperfect numbers come in sets of 10 of the form*

$$\{2 \prod_{i=1}^l p_i, 2^3 \prod_{i=1}^l p_i, 2^7 \prod_{i=1}^l p_i, 2^{15} \prod_{i=1}^l p_i, 2^{31} \prod_{i=1}^l p_i, \\ 2 \prod_{i=1}^{l+1} p_i, 2^3 \prod_{i=1}^{l+1} p_i, 2^7 \prod_{i=1}^{l+1} p_i, 2^{15} \prod_{i=1}^{l+1} p_i, 2^{31} \prod_{i=1}^{l+1} p_i\}$$

where p_l is a Mersenne prime, say $2^p - 1$ and p_{l+1} is the corresponding Wagstaff prime $\frac{2^{p+1}}{3}$.

7.3.3 Basic Pluperfect Numbers

Theorem 7.3.3 *The basic pluperfect numbers are 2^k for $k \in \{1, 3, 7, 15, 31\}$.*

Proof First we investigate the numbers n such that $\phi_Q(n) = 2^k$ for some k . Note that it suffices to understand this on prime powers. So suppose that $2^k = \phi_Q(p^a)$

with some p prime. Then $2^k = p^a - p^{a-1} + \dots + (-1)^a$. Since the left side is even, we have that p and a are both odd. Write $a = 2b + 1$. So

$$2^k = p^{2b+1} - p^{2b-1} + \dots - 1 = (p-1)(p^{2b} + p^{2b-2} + \dots + 1)$$

Write $p-1 = 2^m$ and $p^{2b} + p^{2b-2} + \dots + 1 = 2^n$ with $m+n = k$. Then either $n = 0$ so that $b = 0$, or 2^n is even so that b must be odd. If $b = 0$ then we have $a = 1$ and $p = 2^k + 1$. If not then

$$2^k = (p-1)(p^{2b} + p^{2b-2} + \dots + 1) = (p-1)(p^2 + 1)(p^{2b-2} + p^{2b-6} + \dots + 1)$$

But then $p^2 + 1 = 2^l$, so that $p-1 \mid p^2 + 1$ which is impossible for odd prime p . Therefore we have that if $\phi_Q(n) = 2^k$ then n is a product of primes of the form $2^k + 1$.

Now, suppose that $2^k = \phi_Q(\sigma_P(2^k))$. Then $2^k = \phi_Q(2^{k+1} - 1)$. Then $2^{k+1} - 1$ is a product of primes of the form $2^{2^{n_i}} + 1$ (since all odd primes of the form $2^k + 1$ look like this). Say

$$\prod_{i=1}^l 2^{2^{n_i}} + 1 = 2^{k+1} - 1$$

with all of the n_i distinct. Suppose that $k > 32$. Then we can look at this equation mod 2^2 to get that one of the n_i 's must be 0. Without loss of generality, suppose

it is n_1 . So

$$3 \prod_{i=2}^l 2^{2^{n_i}} + 1 = 2^{k+1} - 1$$

Looking at this equation mod 2^3 we see that one of the n_i 's must be 1. Say $n_2 = 1$.

Then

$$3 * 5 \prod_{i=3}^l 2^{2^{n_i}} + 1 = 2^{k+1} - 1$$

Now looking mod 2^5 we see that, say, $n_3 = 2$. So

$$3 * 5 * 17 \prod_{i=4}^l 2^{2^{n_i}} + 1 = 2^{k+1} - 1$$

Looking mod 2^9 gives $n_4 = 3$ so

$$3 * 5 * 17 * 257 \prod_{i=5}^l 2^{2^{n_i}} + 1 = 2^{k+1} - 1$$

Looking mod 2^{17} we see that $n_5 = 4$ giving

$$3 * 5 * 17 * 257 * 65537 \prod_{i=6}^l 2^{2^{n_i}} + 1 = 2^{k+1} - 1$$

Finally we look at this equation mod 2^{33} . Then we would need $n_6 = 5$, but

$2^{2^5} + 1$ is not prime. Therefore this equation cannot be satisfied with $k > 32$. By

inspection on $k \leq 32$ we have the above result. ■

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